

## MODEL RUN TRANSCRIPT

Generated: 2026-08-12T09:54:58

Source: data/json/noise\_experiments/v2/noise2i\_cap6\_game/claude-sonnet-4.5/game/noisy\_bidirectional\_informed\_memprimary\_cap6/noise2i\_cap6\_game\_000\_neutral\_rep00.json

## RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary_cap6
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 0
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 1
coplayer_history_window: 0
show_agent_names: true
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 0
task_order: [
  "game"
]
agent_names: {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
}
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

## AGENTS

Agent\_1 (Aster); model=anthropic/claude-sonnet-4.5; memory\_capacity=6

### System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent\_2 (Briar); model=anthropic/claude-sonnet-4.5; memory\_capacity=6

#### System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

#### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

### =====

#### ROUND 1

### =====

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

#### Pairings:

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_1 (Aster) sent \$3; received \$9; Agent\_2 (Briar) returned \$2.5; payoffs Agent\_1 (Aster)=\$4.5, Agent\_2 (Briar)=\$6.5; noise/visible: sent visible=\$2.05, received visible=\$6.15, returned visible=\$1.66

Action validation: {"returned": {"amount": 2.5, "clamped": false, "max": 9.0, "min": 0, "raw": 2.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent\_1": 4.5,

"Agent\_2": 6.5

}

Visible balances after round: {

"Agent\_1": 3.66,

"Agent\_2": 3.6499999999999995

```
}
```

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-12T09:51:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}
```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Assistant output:**

Looking at this first round, I need to decide how much trust to place in an unknown receiver.

Since this appears to be a multi-round game, establishing cooperation could be beneficial long-term. Sending money creates value (3x multiplier), but I need the receiver to share the gains fairly.

I'll send a moderate amount that signals trust while managing risk:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return a fair share (around \$4-5), we both benefit compared to not cooperating. This also lets me gauge their trustworthiness for future rounds.

**Reasoning:**

This is the first round of a trust game where I'm playing as the SENDER. Let me think through the strategic considerations:

1. I start with \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned

Key considerations:

- If I send nothing, I keep \$5 and the receiver gets nothing
- If I send everything (\$5), the receiver gets \$15, and then they decide what to return
- There's a note about "communication noise" - amounts may differ slightly from what was actually sent/returned

This is a classic trust game. The optimal strategy depends on:

1. Whether I trust the receiver to reciprocate
2. Whether there will be future rounds with the same person (the prompt mentions "multiple rounds")
3. The communication noise factor

For the first round, I don't have any information about the receiver's behavior. Common approaches:

- Send a moderate amount to signal trust and willingness to cooperate
- See how the receiver responds to gauge their trustworthiness
- In repeated games, cooperation often emerges

A reasonable first move would be to send around \$2-3 to:

- Show willingness to trust and cooperate
- Not risk everything on an unknown partner
- Leave room to adjust strategy based on their response

I'll send \$3 as a fairly trusting opening move that still hedges some risk.

```
Usage: {
  "input_tokens": 365,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

**GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:51:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
```

```

"agent_names": {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
},
"agents": [
  "Agent_1",
  "Agent_2"
],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$2.05, that is 41.0% of their total endowment. You received \$6.149999999999995. How much do you return to The sender? (0-6.149999999999995)

**Assistant output:**

Looking at this situation:

- The sender showed moderate trust by sending \$2.05 (41% of their endowment)
- I received \$6.15 after the 3x multiplier
- The sender currently has \$2.95 remaining

I want to reward their trust while keeping some benefit from being the receiver. I'll return a reasonable proportion that ensures we both profit.

Returning \$2.50 would give:

- Sender:  $\$2.95 + \$2.50 = \$5.45$  (gain of \$0.45)
- Me:  $\$6.15 - \$2.50 = \$3.65$  (gain of \$3.65)

Both of us benefit, and this return (~41% of what I received) mirrors the sender's level of trust.

{'return': 2.50}

#### Reasoning:

Let me analyze this situation:

1. The sender sent \$2.05 out of \$5 (41% of their endowment)
2. This was multiplied by 3x, so I received \$6.15 (approximately)
3. I need to decide how much to return

Key considerations:

- The sender showed moderate trust by sending 41% of their endowment
- If I return nothing, the sender ends up with  $\$5 - \$2.05 = \$2.95$
- If I return a fair amount, we can both benefit
- A common fair strategy is to return enough so both players end up with similar amounts or so the sender at least doesn't lose money

Let me think about fair outcomes:

- Sender currently has:  $\$5 - \$2.05 = \$2.95$
- I currently have: \$6.15
- Total money in system:  $\$2.95 + \$6.15 = \$9.10$

If we split equally: each gets \$4.55

- To achieve this, I'd return:  $\$6.15 - \$4.55 = \$1.60$
- Sender would have:  $\$2.95 + \$1.60 = \$4.55$  ?

Or I could ensure the sender doesn't lose money:

- Sender started with \$5
- Sender sent \$2.05, has \$2.95 left
- To get back to \$5, sender needs:  $\$5 - \$2.95 = \$2.05$
- So I could return \$2.05

Or I could return a proportion that seems fair:

- I received \$6.15
- Returning about 1/3 to 1/2 seems reasonable
- 1/3 of \$6.15 = \$2.05
- 1/2 of \$6.15 = \$3.075

Given that the sender sent 41% (a moderate amount showing some trust), I should return a reasonable amount.

Returning \$2-3 would let both of us profit while rewarding their trust. I'll go with \$2.50, which is roughly 40% of what I received. This gives the sender \$5.45 (a \$0.45 gain) and leaves me with \$3.65 (a \$3.65 gain). The sender took a real risk by sending that much, so returning a meaningful portion acknowledges their willingness to trust....

```
Usage: {
  "input_tokens": 404,
  "output_tokens": 874,
  "reasoning_tokens": 0
}
```

## ROUND 2

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

#### Pairings:

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_2 (Briar) sent \$2; received \$6; Agent\_1 (Aster) returned \$0; payoffs Agent\_2 (Briar)=\$3, Agent\_1 (Aster)=\$6; noise/visible: sent visible=\$0, received visible=\$0  
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

```

Balances after round: {
  "Agent_1": 10.5,
  "Agent_2": 9.5
}
Visible balances after round: {
  "Agent_1": 3.66,
  "Agent_2": 6.6499999999999995
}

```

## LLM CALLS

### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-12T09:51:40  
 Model: anthropic/claude-sonnet-4.5  
 Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may al  
 so be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$2.05, that is 41.0% of their total endowment. You received \$6.1499999999999995. How much do you return to The sender? (0-6.1499999999999995)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed moderate trust by sending \$2.05 (41% of their endowment)
- I received \$6.15 after the 3x multiplier
- The sender currently has \$2.95 remaining

I want to reward their trust while keeping some benefit from being the receiver. I'll return a reasonable proportion that ensures we both profit.

Returning \$2.50 would give:

- Sender:  $\$2.95 + \$2.50 = \$5.45$  (gain of \$0.45)
- Me:  $\$6.15 - \$2.50 = \$3.65$  (gain of \$3.65)

Both of us benefit, and this return (~41% of what I received) mirrors the sender's level of trust.

`{'return': 2.50}`

**reasoning:**

Let me analyze this situation:

1. The sender sent \$2.05 out of \$5 (41% of their endowment)
2. This was multiplied by 3x, so I received \$6.15 (approximately)
3. I need to decide how much to return

Key considerations:

- The sender showed moderate trust by sending 41% of their endowment
- If I return nothing, the sender ends up with  $\$5 - \$2.05 = \$2.95$
- If I return a fair amount, we can both benefit
- A common fair strategy is to return enough so both players end up with similar amounts or so the sender at least doesn't lose money

Let me think about fair outcomes:

- Sender currently has:  $\$5 - \$2.05 = \$2.95$
- I currently have: \$6.15
- Total money in system:  $\$2.95 + \$6.15 = \$9.10$

If we split equally: each gets \$4.55

- To achieve this, I'd return:  $\$6.15 - \$4.55 = \$1.60$
- Sender would have:  $\$2.95 + \$1.60 = \$4.55$  ?

Or I could ensure the sender doesn't lose money:

- Sender started with \$5
- Sender sent \$2.05, has \$2.95 left
- To get back to \$5, sender needs:  $\$5 - \$2.95 = \$2.05$
- So I could return \$2.05

Or I could return a proportion that seems fair:

- I received \$6.15
- Returning about 1/3 to 1/2 seems reasonable
- 1/3 of \$6.15 = \$2.05
- 1/2 of \$6.15 = \$3.075

Given that the sender sent 41% (a moderate amount showing some trust), I should return a reasonable amount.



Returning \$2-3 would let both of us profit while rewarding their trust. I'll go with \$2.50, which is roughly 40% of what I received. This gives the sender \$5.45 (a \$0.45 gain) and leaves me with \$3.65 (a \$3.65 gain). The sender took a real risk by sending that much, so returning a meaningful portion acknowledges their willingness to trust....

**usage:**

```
{
  "input_tokens": 404,
  "output_tokens": 874,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.6499999999999995.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at Round 1, the sender showed moderate trust by sending 41% of their endowment, and I reciprocated with a fair return. This established a cooperative pattern.

Now as the sender, I'll maintain this level of trust and cooperation. Sending a moderate amount signals trust while managing risk.

I'll send \$2.00 (40% of my endowment):

- The receiver will get \$6.00 after the multiplier
- If they reciprocate similarly to how I did in Round 1, both of us should benefit
- This maintains the cooperative dynamic established in Round 1

```
{'send': 2.00}
```

**Reasoning:**

Okay, now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was the receiver:

- Sender sent me \$2.05 (41% of their endowment)
- I received \$6.15
- I returned \$2.50
- So the sender ended up with  $\$2.95 + \$2.50 = \$5.45$
- I ended up with  $\$6.15 - \$2.50 = \$3.65$

The sender in Round 1 showed moderate trust by sending 41% of their endowment, and I reciprocated with a return that was about 41% of what I received. This established a pattern of moderate cooperation.

Now I'm the sender. I need to decide how much to send.

Key considerations:

1. The previous sender sent 41% and got a decent return
2. I should signal trust but also be strategic
3. There's communication noise mentioned, so amounts may differ from what's actually sent
4. I want to maximize my total earnings

If I send \$2 (40%):

- Receiver gets \$6
- If they return similarly to how I did (~41%), they might return \$2.50
- I'd have:  $\$3 + \$2.50 = \$5.50$

If I send \$2.50 (50%):

- Receiver gets \$7.50
- If they return ~40%, they might return \$3
- I'd have:  $\$2.50 + \$3 = \$5.50$

If I send \$3 (60%):

- Receiver gets \$9
- If they return ~40%, they might return \$3.60
- I'd have:  $\$2 + \$3.60 = \$5.60$

If I send \$1.50 (30%):

...

```
Usage: {
  "input_tokens": 644,
  "output_tokens": 691,
  "reasoning_tokens": 0
}
```

**GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:51:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to decide how much trust to place in an unknown receiver.

Since this appears to be a multi-round game, establishing cooperation could be beneficial long-term. Sending money creates value (3x multiplier), but I need the receiver to share the gains fairly.

I'll send a moderate amount that signals trust while managing risk:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return a fair share (around \$4-5), we both benefit compared to not cooperating. This also lets me gauge their trustworthiness for future rounds.

**reasoning:**

This is the first round of a trust game where I'm playing as the SENDER. Let me think through the strategic considerations:

1. I start with \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned

Key considerations:

- If I send nothing, I keep \$5 and the receiver gets nothing
- If I send everything (\$5), the receiver gets \$15, and then they decide what to return
- There's a note about "communication noise" - amounts may differ slightly from what was actually sent/returned

This is a classic trust game. The optimal strategy depends on:

1. Whether I trust the receiver to reciprocate
2. Whether there will be future rounds with the same person (the prompt mentions "multiple rounds")
3. The communication noise factor

For the first round, I don't have any information about the receiver's behavior. Common approaches:

- Send a moderate amount to signal trust and willingness to cooperate
- See how the receiver responds to gauge their trustworthiness
- In repeated games, cooperation often emerges

A reasonable first move would be to send around \$2-3 to:

- Show willingness to trust and cooperate
- Not risk everything on an unknown partner
- Leave room to adjust strategy based on their response

I'll send \$3 as a fairly trusting opening move that still hedges some risk.

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$3.66.  
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender?

er? (0-0)

**Assistant output:**

Since the sender sent me \$0, I received \$0 (even after the 3x multiplier).

With \$0 received, I have nothing to return. The only possible amount is:

```
'''json
{"return": 0}
'''
```

**Reasoning:**

Let me understand the situation:

Round 1: I was SENDER and sent \$3

- I kept:  $\$5 - \$3 = \$2$
- Receiver got:  $\$3 \times 3 = \$9$
- My total earnings are \$3.66

This means I must have received \$1.66 back from the receiver in Round 1.

So:  $\$2$  (kept) +  $\$1.66$  (returned) =  $\$3.66$  ?

Round 2: I am RECEIVER

- The sender sent me \$0
- I received \$0 (since  $\$0 \times 3 = \$0$ )
- I can only return between \$0 and \$0

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

```
Usage: {
  "input_tokens": 567,
  "output_tokens": 245,
  "reasoning_tokens": 0
}
```

=====

**ROUND 3**

=====

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

**Pairings:**

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_1 (Aster) sent \$2; received \$6; Agent\_2 (Briar) returned \$0; payoffs Agent\_1

(Aster)=\$3, Agent\_2 (Briar)=\$6; noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent\_1": 13.5,  
"Agent\_2": 15.5

}

Visible balances after round: {

"Agent\_1": 6.66,  
"Agent\_2": 6.6499999999999995

}

**LLM CALLS**

**GAME - Agent\_1 - Sender - dyad\_1**

Timestamp: 2026-08-12T09:51:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_2",  
"pairing": {  
 "agent\_names": {  
 "Agent\_1": "Aster",  
 "Agent\_2": "Briar"

},

"agents": [  
 "Agent\_1",

```

    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to decide how much trust to place in an unknown receiver.

Since this appears to be a multi-round game, establishing cooperation could be beneficial long-term. Sending money creates value (3x multiplier), but I need the receiver to share the gains fairly.

I'll send a moderate amount that signals trust while managing risk:

```

'''json
{"send": 3}
'''

```

This sends \$3, which becomes \$9 for the receiver. If they return a fair share (around \$4-5), we both benefit compared to not cooperating. This also lets me gauge their trustworthiness for future rounds.

**reasoning:**

This is the first round of a trust game where I'm playing as the SENDER. Let me think through the strategic considerations:

1. I start with \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned

**Key considerations:**

- If I send nothing, I keep \$5 and the receiver gets nothing
- If I send everything (\$5), the receiver gets \$15, and then they decide what to return
- There's a note about "communication noise" - amounts may differ slightly from what was actually sent/returned

This is a classic trust game. The optimal strategy depends on:

1. Whether I trust the receiver to reciprocate
2. Whether there will be future rounds with the same person (the prompt mentions "multiple rounds")
3. The communication noise factor

For the first round, I don't have any information about the receiver's behavior. Common approaches:

- Send a moderate amount to signal trust and willingness to cooperate
- See how the receiver responds to gauge their trustworthiness
- In repeated games, cooperation often emerges

A reasonable first move would be to send around \$2-3 to:

- Show willingness to trust and cooperate
- Not risk everything on an unknown partner
- Leave room to adjust strategy based on their response

I'll send \$3 as a fairly trusting opening move that still hedges some risk.

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$3.66.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

Since the sender sent me \$0, I received \$0 (even after the 3x multiplier).

With \$0 received, I have nothing to return. The only possible amount is:

```
'''json
{"return": 0}
'''
```

**reasoning:**

Let me understand the situation:

Round 1: I was SENDER and sent \$3

- I kept: \$5 - \$3 = \$2
- Receiver got: \$3 × 3 = \$9
- My total earnings are \$3.66

This means I must have received \$1.66 back from the receiver in Round 1.

So: \$2 (kept) + \$1.66 (returned) = \$3.66 ?

Round 2: I am RECEIVER

- The sender sent me \$0
- I received \$0 (since  $\$0 \times 3 = \$0$ )
- I can only return between \$0 and \$0

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

**usage:**

```
{
  "input_tokens": 567,
  "output_tokens": 245,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$3.66.  
How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my situation:

From Round 1, I can calculate that the receiver only returned \$1.66 out of the \$9 they received (about 18% return rate), which shows low reciprocity. However, I should note the warning about communication noise - the actual amounts may have differed.

Round 2 didn't affect my earnings since the sender sent \$0.

For Round 3, I'm facing uncertainty:

- This could be a different receiver
- Communication noise may have affected Round 1 observations
- I need to balance potential gains against the risk of poor reciprocity

I'll send a smaller amount to test this receiver's trustworthiness while limiting my downside risk:

```
'''json
{"send": 2}
'''
```

This sends \$2, which becomes \$6 for the receiver. It's enough to create mutual benefit if they reciprocate fairly, but limits my loss if they don't.

**Reasoning:**

Let me analyze the situation:

Round 1: I was the SENDER and sent \$3

- My total earnings are now \$3.66
- If I sent \$3, I kept  $\$5 - \$3 = \$2$
- The receiver got  $\$3 \times 3 = \$9$
- They returned some amount to me
- So I have:  $\$2 + \text{returned amount} = \$3.66$
- This means they returned \$1.66 to me out of the \$9 they received

That's a pretty poor return rate ( $1.66/9 \approx 18.4\%$ ). They kept \$7.34 out of \$9.

Round 2: I was the RECEIVER

- The sender sent me \$0
- So I received \$0 and returned \$0
- This didn't affect my earnings

Now in Round 3, I'm the SENDER again.

The receiver from Round 1 showed low trustworthiness by only returning \$1.66 out of \$9. However, I should note that the instructions mention "communication noise" - the amounts I see may differ from what was actually sent/returned.

Given that:

1. The previous receiver showed poor reciprocity (only returned ~18% of what they received)
2. There may be communication noise affecting what I observe
3. This might be a different receiver (it's a multi-round game, roles change)

I have a few options:

- Send \$0 to protect my earnings (risk-averse)
- Send a small amount to test this receiver
- Send a moderate amount hoping for better reciprocity

Given the poor experience in Round 1, I'll be more cautious. I'll send a smaller amount to test this receiver's trustworthiness:

I'll send \$2 this time.

```
Usage: {
  "input_tokens": 668,
  "output_tokens": 617,
  "reasoning_tokens": 0
}
```

#### GAME - Agent\_2 - Receiver - dyad\_1

Timestamp: 2026-08-12T09:52:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned



- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$2.05, that is 41.0% of their total endowment. You received \$6.1499999999999995. How much do you return to The sender? (0-6.1499999999999995)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed moderate trust by sending \$2.05 (41% of their endowment)
- I received \$6.15 after the 3x multiplier
- The sender currently has \$2.95 remaining

I want to reward their trust while keeping some benefit from being the receiver. I'll return a reasonable proportion that ensures we both profit.

Returning \$2.50 would give:

- Sender:  $\$2.95 + \$2.50 = \$5.45$  (gain of \$0.45)
- Me:  $\$6.15 - \$2.50 = \$3.65$  (gain of \$3.65)

Both of us benefit, and this return (~41% of what I received) mirrors the sender's level of trust.

{'return': 2.50}

**reasoning:**

Let me analyze this situation:

1. The sender sent \$2.05 out of \$5 (41% of their endowment)
2. This was multiplied by 3x, so I received \$6.15 (approximately)
3. I need to decide how much to return

Key considerations:

- The sender showed moderate trust by sending 41% of their endowment
- If I return nothing, the sender ends up with  $\$5 - \$2.05 = \$2.95$
- If I return a fair amount, we can both benefit
- A common fair strategy is to return enough so both players end up with similar amounts or so the sender at least doesn't lose money

Let me think about fair outcomes:

- Sender currently has:  $\$5 - \$2.05 = \$2.95$
- I currently have: \$6.15
- Total money in system:  $\$2.95 + \$6.15 = \$9.10$

If we split equally: each gets \$4.55

- To achieve this, I'd return:  $\$6.15 - \$4.55 = \$1.60$
- Sender would have:  $\$2.95 + \$1.60 = \$4.55$  ?

Or I could ensure the sender doesn't lose money:

- Sender started with \$5
- Sender sent \$2.05, has \$2.95 left
- To get back to \$5, sender needs:  $\$5 - \$2.95 = \$2.05$
- So I could return \$2.05

Or I could return a proportion that seems fair:

- I received \$6.15
- Returning about 1/3 to 1/2 seems reasonable
- 1/3 of \$6.15 = \$2.05

-  $1/2$  of \$6.15 = \$3.075

Given that the sender sent 41% (a moderate amount showing some trust), I should return a reasonable amount.

Returning \$2-3 would let both of us profit while rewarding their trust. I'll go with \$2.50, which is roughly 40% of what I received. This gives the sender \$5.45 (a \$0.45 gain) and leaves me with \$3.65 (a \$3.65 gain). The sender took a real risk by sending that much, so returning a meaningful portion acknowledges their willingness to trust....

**usage:**

```
{
  "input_tokens": 404,
  "output_tokens": 874,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.6499999999999995.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at Round 1, the sender showed moderate trust by sending 41% of their endowment, and I reciprocated with a fair return. This established a cooperative pattern.

Now as the sender, I'll maintain this level of trust and cooperation. Sending a moderate amount signals trust while managing risk.

I'll send \$2.00 (40% of my endowment):

- The receiver will get \$6.00 after the multiplier
- If they reciprocate similarly to how I did in Round 1, both of us should benefit
- This maintains the cooperative dynamic established in Round 1

{'send': 2.00}

**reasoning:**

Okay, now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was the receiver:

- Sender sent me \$2.05 (41% of their endowment)
- I received \$6.15
- I returned \$2.50
- So the sender ended up with  $\$2.95 + \$2.50 = \$5.45$
- I ended up with  $\$6.15 - \$2.50 = \$3.65$

The sender in Round 1 showed moderate trust by sending 41% of their endowment, and I reciprocated with a return that was about 41% of what I received. This established a pattern of moderate cooperation.

Now I'm the sender. I need to decide how much to send.

Key considerations:

1. The previous sender sent 41% and got a decent return
2. I should signal trust but also be strategic
3. There's communication noise mentioned, so amounts may differ from what's actually sent
4. I want to maximize my total earnings

If I send \$2 (40%):

- Receiver gets \$6
- If they return similarly to how I did (~41%), they might return \$2.50
- I'd have:  $\$3 + \$2.50 = \$5.50$

If I send \$2.50 (50%):

- Receiver gets \$7.50
- If they return ~40%, they might return \$3
- I'd have:  $\$2.50 + \$3 = \$5.50$

If I send \$3 (60%):

- Receiver gets \$9

- If they return ~40%, they might return \$3.60
- I'd have: \$2 + \$3.60 = \$5.60

If I send \$1.50 (30%):

...

**usage:**

```
{
  "input_tokens": 644,
  "output_tokens": 691,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.6499999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

The sender sent \$0 this round, so I received \$0. Since I have nothing to return, my only option is:

```
{'return': 0}
```

**Reasoning:**

The sender sent me \$0 this round, so I received \$0 after multiplication.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust me
2. They're being strategic
3. There might be communication noise mentioned in the instructions

Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

This is straightforward - there's nothing to return.

Usage: {

```
"input_tokens": 851,
"output_tokens": 148,
"reasoning_tokens": 0
}
```

## ROUND 4

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

**Pairings:**

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_2 (Briar) sent \$1.5; received \$4.5; Agent\_1 (Aster) returned \$1.5; payoffs Agent\_2 (Briar)=\$5, Agent\_1 (Aster)=\$3; noise/visible: sent visible=\$0.98, received visible=\$2.94, returned visible=\$1.21

Action validation: {"returned": {"amount": 1.5, "clamped": false, "max": 4.5, "min": 0, "raw": 1.5}, "sent": {"amount": 1.5, "clamped": false, "max": 5, "min": 0, "raw": 1.5}}

Balances after round: {

```
"Agent_1": 16.5,
"Agent_2": 20.5
}
```

Visible balances after round: {

```
"Agent_1": 8.1,
"Agent_2": 11.36
}
```

## LLM CALLS

**GAME - Agent\_2 - Sender - dyad\_1**

Timestamp: 2026-08-12T09:52:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
```

```

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$2.05, that is 41.0% of their total endowment. You received \$6.149999999999995. How much do you return to The sender? (0-6.149999999999995)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed moderate trust by sending \$2.05 (41% of their endowment)
- I received \$6.15 after the 3x multiplier
- The sender currently has \$2.95 remaining

I want to reward their trust while keeping some benefit from being the receiver. I'll return a reasonable proportion that ensures we both profit.

Returning \$2.50 would give:

- Sender:  $\$2.95 + \$2.50 = \$5.45$  (gain of \$0.45)
- Me:  $\$6.15 - \$2.50 = \$3.65$  (gain of \$3.65)

Both of us benefit, and this return (~41% of what I received) mirrors the sender's level of trust.

```
{'return': 2.50}
```

#### reasoning:

Let me analyze this situation:

1. The sender sent \$2.05 out of \$5 (41% of their endowment)
2. This was multiplied by 3x, so I received \$6.15 (approximately)
3. I need to decide how much to return

Key considerations:

- The sender showed moderate trust by sending 41% of their endowment
- If I return nothing, the sender ends up with  $\$5 - \$2.05 = \$2.95$
- If I return a fair amount, we can both benefit
- A common fair strategy is to return enough so both players end up with similar amounts or so the sender at least doesn't lose money

Let me think about fair outcomes:

- Sender currently has:  $\$5 - \$2.05 = \$2.95$
- I currently have: \$6.15
- Total money in system:  $\$2.95 + \$6.15 = \$9.10$

If we split equally: each gets \$4.55

- To achieve this, I'd return:  $\$6.15 - \$4.55 = \$1.60$
- Sender would have:  $\$2.95 + \$1.60 = \$4.55$  ?

Or I could ensure the sender doesn't lose money:

- Sender started with \$5
- Sender sent \$2.05, has \$2.95 left
- To get back to \$5, sender needs:  $\$5 - \$2.95 = \$2.05$
- So I could return \$2.05

Or I could return a proportion that seems fair:

- I received \$6.15
- Returning about 1/3 to 1/2 seems reasonable
- 1/3 of \$6.15 = \$2.05
- 1/2 of \$6.15 = \$3.075

Given that the sender sent 41% (a moderate amount showing some trust), I should return a reasonable amount.

Returning \$2-3 would let both of us profit while rewarding their trust. I'll go with \$2.50, which is roughly 40% of what I received. This gives the sender \$5.45 (a \$0.45 gain) and leaves me with \$3.65 (a \$3.65 gain). The sender took a real risk by sending that much, so returning a meaningful portion acknowledges their willingness to trust....

#### usage:

```
{
  "input_tokens": 404,
  "output_tokens": 874,
  "reasoning_tokens": 0
}
```

#### Message 4 role=user

##### content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.6499999999999995.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at Round 1, the sender showed moderate trust by sending 41% of their endowment, and I reciprocated with a fair return. This established a cooperative pattern.

Now as the sender, I'll maintain this level of trust and cooperation. Sending a moderate amount signals trust while managing risk.

I'll send \$2.00 (40% of my endowment):

- The receiver will get \$6.00 after the multiplier
- If they reciprocate similarly to how I did in Round 1, both of us should benefit
- This maintains the cooperative dynamic established in Round 1

{'send': 2.00}

**reasoning:**

Okay, now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was the receiver:

- Sender sent me \$2.05 (41% of their endowment)
- I received \$6.15
- I returned \$2.50
- So the sender ended up with  $\$2.95 + \$2.50 = \$5.45$
- I ended up with  $\$6.15 - \$2.50 = \$3.65$

The sender in Round 1 showed moderate trust by sending 41% of their endowment, and I reciprocated with a return that was about 41% of what I received. This established a pattern of moderate cooperation.

Now I'm the sender. I need to decide how much to send.

Key considerations:

1. The previous sender sent 41% and got a decent return
2. I should signal trust but also be strategic
3. There's communication noise mentioned, so amounts may differ from what's actually sent
4. I want to maximize my total earnings

If I send \$2 (40%):

- Receiver gets \$6
- If they return similarly to how I did (~41%), they might return \$2.50
- I'd have:  $\$3 + \$2.50 = \$5.50$

If I send \$2.50 (50%):

- Receiver gets \$7.50
- If they return ~40%, they might return \$3
- I'd have:  $\$2.50 + \$3 = \$5.50$

If I send \$3 (60%):

- Receiver gets \$9
- If they return ~40%, they might return \$3.60
- I'd have:  $\$2 + \$3.60 = \$5.60$

If I send \$1.50 (30%):

...

**usage:**

```
{
  "input_tokens": 644,
  "output_tokens": 691,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.6499999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 7 role=assistant**

**content:**

The sender sent \$0 this round, so I received \$0. Since I have nothing to return, my only option is:

```
{'return': 0}
```

**reasoning:**

The sender sent me \$0 this round, so I received \$0 after multiplication.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust me
2. They're being strategic
3. There might be communication noise mentioned in the instructions

Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

This is straightforward - there's nothing to return.

**usage:**

```
{
  "input_tokens": 851,
  "output_tokens": 148,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.6499999999999995.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the pattern so far:

- Round 1: I cooperated well as receiver, returning \$2.50
- Round 2: I sent \$2.00 but apparently received nothing back (my earnings suggest \$0 return)
- Round 3: Sender sent me \$0

There's been a breakdown in cooperation, though the note about communication noise means perceptions might differ from reality.

I'll send a moderate amount to signal willingness to cooperate while being cautious given recent experiences. Sending \$1.50 allows me to:

- Test if cooperation can be rebuilt
- Limit potential losses if trust isn't reciprocated
- Keep open the possibility of mutual benefit

```
{'send': 1.50}
```

**Reasoning:**

Let me review what's happened:

Round 1: I was receiver, got \$6.15, returned \$2.50

Round 2: I was sender, sent \$2.00

Round 3: I was receiver, sender sent \$0, I returned \$0

My total earnings are \$6.65.

Round 3 is concerning - the sender sent me \$0. This could mean:

1. They didn't trust me to return anything
2. They're being cautious or strategic
3. There might be communication noise affecting their perception
4. They're testing cooperation levels

The instructions mentioned there may be communication noise - amounts might differ from what was actually sent/returned.

In Round 2, I sent \$2.00. I don't know what they returned yet (that's not shown in my earnings update for round 3).

Wait, let me recalculate my earnings:

- Round 1: Started with \$0, received \$6.15, returned \$2.50, kept \$3.65
- Round 2: Started with \$5, sent \$2.00, kept \$3.00 initially. Need to add what was returned to me.

- Round 3: Received \$0, returned \$0, gained \$0

Total earnings = \$6.65

From Round 1: \$3.65

From Round 3: \$0

So from Round 2: \$6.65 - \$3.65 - \$0 = \$3.00

This means in Round 2, I sent \$2.00 and got back \$0. The receiver didn't return anything!

So the pattern is:

- Round 1: Sender sent me \$2.05, I returned \$2.50 (good cooperation from me)
- Round 2: I sent \$2.00, receiver returned \$0 (betrayal!)
- Round 3: Sender sent me \$0 (maybe retaliating? or cautious?)

The zero in Round 3 could be retaliation for my Round 2 behavior, or it might stem from communication distortion about what I actually returned in Round 1. Since each round likely involves different partners, the person who wit...

```
Usage: {  
  "input_tokens": 937,  
  "output_tokens": 851,  
  "reasoning_tokens": 0  
}
```

#### GAME - Agent\_1 - Receiver - dyad\_1

Timestamp: 2026-08-12T09:52:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned



- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to decide how much trust to place in an unknown receiver.

Since this appears to be a multi-round game, establishing cooperation could be beneficial long-term. Sending money creates value (3x multiplier), but I need the receiver to share the gains fairly.

I'll send a moderate amount that signals trust while managing risk:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return a fair share (around \$4-5), we both benefit compared to not cooperating. This also lets me gauge their trustworthiness for future rounds.

**reasoning:**

This is the first round of a trust game where I'm playing as the SENDER. Let me think through the strategic considerations:

1. I start with \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned

Key considerations:

- If I send nothing, I keep \$5 and the receiver gets nothing
- If I send everything (\$5), the receiver gets \$15, and then they decide what to return
- There's a note about "communication noise" - amounts may differ slightly from what was actually sent/returned

This is a classic trust game. The optimal strategy depends on:

1. Whether I trust the receiver to reciprocate
2. Whether there will be future rounds with the same person (the prompt mentions "multiple rounds")
3. The communication noise factor

For the first round, I don't have any information about the receiver's behavior. Common approaches:

- Send a moderate amount to signal trust and willingness to cooperate
- See how the receiver responds to gauge their trustworthiness
- In repeated games, cooperation often emerges

A reasonable first move would be to send around \$2-3 to:

- Show willingness to trust and cooperate
- Not risk everything on an unknown partner
- Leave room to adjust strategy based on their response

I'll send \$3 as a fairly trusting opening move that still hedges some risk.

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$3.66.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

Since the sender sent me \$0, I received \$0 (even after the 3x multiplier).

With \$0 received, I have nothing to return. The only possible amount is:

```
'''json
{"return": 0}
'''
```

**reasoning:**

Let me understand the situation:

Round 1: I was SENDER and sent \$3

- I kept:  $\$5 - \$3 = \$2$
- Receiver got:  $\$3 \times 3 = \$9$
- My total earnings are \$3.66

This means I must have received \$1.66 back from the receiver in Round 1.

So:  $\$2$  (kept) +  $\$1.66$  (returned) =  $\$3.66$  ?

Round 2: I am RECEIVER

- The sender sent me \$0
- I received \$0 (since  $\$0 \times 3 = \$0$ )
- I can only return between \$0 and \$0

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

**usage:**

```
{
  "input_tokens": 567,
  "output_tokens": 245,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$3.66.

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at my situation:

From Round 1, I can calculate that the receiver only returned \$1.66 out of the \$9 they received (about 18% return rate), which shows low reciprocity. However, I should note the warning about communication noise - the actual amounts may have differed.

Round 2 didn't affect my earnings since the sender sent \$0.

For Round 3, I'm facing uncertainty:

- This could be a different receiver
- Communication noise may have affected Round 1 observations

- I need to balance potential gains against the risk of poor reciprocity

I'll send a smaller amount to test this receiver's trustworthiness while limiting my downside risk:

```
'''json
{"send": 2}
'''
```

This sends \$2, which becomes \$6 for the receiver. It's enough to create mutual benefit if they reciprocate fairly, but limits my loss if they don't.

**reasoning:**

Let me analyze the situation:

Round 1: I was the SENDER and sent \$3

- My total earnings are now \$3.66
- If I sent \$3, I kept \$5 - \$3 = \$2
- The receiver got  $\$3 \times 3 = \$9$
- They returned some amount to me
- So I have:  $\$2 + \text{returned amount} = \$3.66$
- This means they returned \$1.66 to me out of the \$9 they received

That's a pretty poor return rate ( $1.66/9 \approx 18.4\%$ ). They kept \$7.34 out of \$9.

Round 2: I was the RECEIVER

- The sender sent me \$0
- So I received \$0 and returned \$0
- This didn't affect my earnings

Now in Round 3, I'm the SENDER again.

The receiver from Round 1 showed low trustworthiness by only returning \$1.66 out of \$9. However, I should note that the instructions mention "communication noise" - the amounts I see may differ from what was actually sent/returned.

Given that:

1. The previous receiver showed poor reciprocity (only returned ~18% of what they received)
2. There may be communication noise affecting what I observe
3. This might be a different receiver (it's a multi-round game, roles change)

I have a few options:

- Send \$0 to protect my earnings (risk-averse)
- Send a small amount to test this receiver
- Send a moderate amount hoping for better reciprocity

Given the poor experience in Round 1, I'll be more cautious. I'll send a smaller amount to test this receiver's trustworthiness:

I'll send \$2 this time.

**usage:**

```
{
  "input_tokens": 668,
  "output_tokens": 617,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$6.66.

This round, The sender sent you \$0.98, so you received \$2.94. How much do you return to the sender? (0-2.94)

**Assistant output:**

Looking at this situation:

The sender sent me \$0.98, which became \$2.94 after the 3x multiplier. This created value through cooperation.

To encourage continued cooperation and show trustworthiness, I should return a fair amount. Let

- Let me consider the gains:
- Total value created from cooperation: \$1.96 (\$2.94 - \$0.98)
  - A fair return would split these gains reasonably

If I return \$1.50:

- Sender gains: \$5 - \$0.98 + \$1.50 = \$5.52 (net gain: \$0.52)
- I gain: \$2.94 - \$1.50 = \$1.44 (net gain: \$1.44)

This gives us both positive gains from cooperating, with me gaining slightly more since I received the multiplied amount.

```
'''json
{"return": 1.50}
'''
```

#### Reasoning:

Let me analyze this situation:

The sender sent me \$0.98, which was multiplied by 3x to give me \$2.94.

I need to decide how much to return. Let me think about fairness and cooperation:

1. The sender gave up \$0.98 from their \$5 endowment
2. This created \$2.94 for me
3. The total value created is \$2.94 - \$0.98 = \$1.96 in additional value due to the multiplier

A fair split might consider:

- Equal split of gains: The sender sacrificed \$0.98, I got \$2.94. If I return nothing, sender has \$4.02 and I have \$2.94. If we split the \$1.96 gain equally, I should return enough so we each gain about \$0.98.
- To give the sender a total gain of ~\$0.98, they need: \$5 - \$0.98 + return = \$5.98, so return = \$0.98
- That seems too low. Let me think differently.

Alternative approach - equal final outcomes:

- Sender started with \$5, spent \$0.98, has \$4.02 + return
- I started with \$0, got \$2.94, keep \$2.94 - return
- For equal outcomes: \$4.02 + return = \$2.94 - return
- 2\*return = \$2.94 - \$4.02 = -\$1.08
- This gives negative, which doesn't make sense

Let me think about reciprocity:

- The sender trusted me with \$0.98
- This created value of \$1.96 (the difference between 3x multiplier and original)
- A reasonable return might be to give back enough that we both benefit from the cooperation
- Testing different return amounts: returning \$1.47 would give us nearly equal final gains (\$0.49 for them, \$1.47 for me), while \$1.50 provides them \$0.52 in gains. Even \$1.20 leaves them with only \$0.22 in gains while I keep \$1.74....

```
Usage: {
  "input_tokens": 938,
  "output_tokens": 1003,
  "reasoning_tokens": 0
}
```

#### ROUND 5

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

#### Pairings:

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_1 (Aster) sent \$1; received \$3; Agent\_2 (Briar) returned \$0; payoffs Agent\_1 (Aster)=\$4, Agent\_2 (Briar)=\$3; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.81

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 3.0, "min": 0, "raw": 0.0}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

"Agent\_1": 20.5,  
"Agent\_2": 23.5

}

Visible balances after round: {

```

"Agent_1": 12.91,
"Agent_2": 11.36
}

```

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-12T09:53:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}

```

### Exact input messages sent to model:

Message 1 role=system

#### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

#### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to decide how much trust to place in an unknown receiver.

Since this appears to be a multi-round game, establishing cooperation could be beneficial long-term. Sending money creates value (3x multiplier), but I need the receiver to share the gains fairly.

I'll send a moderate amount that signals trust while managing risk:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return a fair share (around \$4-5), we both benefit compared to not cooperating. This also lets me gauge their trustworthiness for future rounds.

**reasoning:**

This is the first round of a trust game where I'm playing as the SENDER. Let me think through the strategic considerations:

1. I start with \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned

Key considerations:

- If I send nothing, I keep \$5 and the receiver gets nothing
- If I send everything (\$5), the receiver gets \$15, and then they decide what to return
- There's a note about "communication noise" - amounts may differ slightly from what was actually sent/returned

This is a classic trust game. The optimal strategy depends on:

1. Whether I trust the receiver to reciprocate
2. Whether there will be future rounds with the same person (the prompt mentions "multiple rounds")
3. The communication noise factor

For the first round, I don't have any information about the receiver's behavior. Common approaches:

- Send a moderate amount to signal trust and willingness to cooperate
- See how the receiver responds to gauge their trustworthiness
- In repeated games, cooperation often emerges

A reasonable first move would be to send around \$2-3 to:

- Show willingness to trust and cooperate
- Not risk everything on an unknown partner
- Leave room to adjust strategy based on their response

I'll send \$3 as a fairly trusting opening move that still hedges some risk.

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$3.66.  
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender?

er? (0-0)

**Message 5 role=assistant**

**content:**

Since the sender sent me \$0, I received \$0 (even after the 3x multiplier).

With \$0 received, I have nothing to return. The only possible amount is:

```
'''json
{"return": 0}
'''
```

**reasoning:**

Let me understand the situation:

Round 1: I was SENDER and sent \$3

- I kept:  $\$5 - \$3 = \$2$
- Receiver got:  $\$3 \times 3 = \$9$
- My total earnings are \$3.66

This means I must have received \$1.66 back from the receiver in Round 1.

So:  $\$2$  (kept) +  $\$1.66$  (returned) =  $\$3.66$  ?

Round 2: I am RECEIVER

- The sender sent me \$0
- I received \$0 (since  $\$0 \times 3 = \$0$ )
- I can only return between \$0 and \$0

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

**usage:**

```
{
  "input_tokens": 567,
  "output_tokens": 245,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$3.66.

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at my situation:

From Round 1, I can calculate that the receiver only returned \$1.66 out of the \$9 they received (about 18% return rate), which shows low reciprocity. However, I should note the warning about communication noise - the actual amounts may have differed.

Round 2 didn't affect my earnings since the sender sent \$0.

For Round 3, I'm facing uncertainty:

- This could be a different receiver
- Communication noise may have affected Round 1 observations
- I need to balance potential gains against the risk of poor reciprocity

I'll send a smaller amount to test this receiver's trustworthiness while limiting my downside risk:

```
'''json
{"send": 2}
'''
```

This sends \$2, which becomes \$6 for the receiver. It's enough to create mutual benefit if they reciprocate fairly, but limits my loss if they don't.

**reasoning:**

Let me analyze the situation:

Round 1: I was the SENDER and sent \$3

- My total earnings are now \$3.66
- If I sent \$3, I kept  $\$5 - \$3 = \$2$

- The receiver got  $\$3 \times 3 = \$9$
- They returned some amount to me
- So I have:  $\$2 + \text{returned amount} = \$3.66$
- This means they returned  $\$1.66$  to me out of the  $\$9$  they received

That's a pretty poor return rate ( $1.66/9 \approx 18.4\%$ ). They kept  $\$7.34$  out of  $\$9$ .

Round 2: I was the RECEIVER

- The sender sent me  $\$0$
- So I received  $\$0$  and returned  $\$0$
- This didn't affect my earnings

Now in Round 3, I'm the SENDER again.

The receiver from Round 1 showed low trustworthiness by only returning  $\$1.66$  out of  $\$9$ . However, I should note that the instructions mention "communication noise" - the amounts I see may differ from what was actually sent/returned.

Given that:

1. The previous receiver showed poor reciprocity (only returned  $\sim 18\%$  of what they received)
2. There may be communication noise affecting what I observe
3. This might be a different receiver (it's a multi-round game, roles change)

I have a few options:

- Send  $\$0$  to protect my earnings (risk-averse)
- Send a small amount to test this receiver
- Send a moderate amount hoping for better reciprocity

Given the poor experience in Round 1, I'll be more cautious. I'll send a smaller amount to test this receiver's trustworthiness:

I'll send  $\$2$  this time.

**usage:**

```
{
  "input_tokens": 668,
  "output_tokens": 617,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are  $\$6.66$ .

This round, The sender sent you  $\$0.98$ , so you received  $\$2.94$ . How much do you return to The sender? (0-2.94)

**Message 9 role=assistant**

**content:**

Looking at this situation:

The sender sent me  $\$0.98$ , which became  $\$2.94$  after the 3x multiplier. This created value through cooperation.

To encourage continued cooperation and show trustworthiness, I should return a fair amount. Let me consider the gains:

- Total value created from cooperation:  $\$1.96$  ( $\$2.94 - \$0.98$ )
- A fair return would split these gains reasonably

If I return  $\$1.50$ :

- Sender gains:  $\$5 - \$0.98 + \$1.50 = \$5.52$  (net gain:  $\$0.52$ )
- I gain:  $\$2.94 - \$1.50 = \$1.44$  (net gain:  $\$1.44$ )

This gives us both positive gains from cooperating, with me gaining slightly more since I received the multiplied amount.

```
'''json
{"return": 1.50}
'''
```

**reasoning:**

Let me analyze this situation:



The sender sent me \$0.98, which was multiplied by 3x to give me \$2.94.

I need to decide how much to return. Let me think about fairness and cooperation:

1. The sender gave up \$0.98 from their \$5 endowment
2. This created \$2.94 for me
3. The total value created is  $\$2.94 - \$0.98 = \$1.96$  in additional value due to the multiplier

A fair split might consider:

- Equal split of gains: The sender sacrificed \$0.98, I got \$2.94. If I return nothing, sender has \$4.02 and I have \$2.94. If we split the \$1.96 gain equally, I should return enough so we each gain about \$0.98.
- To give the sender a total gain of ~\$0.98, they need:  $\$5 - \$0.98 + \text{return} = \$5.98$ , so return = \$0.98
- That seems too low. Let me think differently.

Alternative approach - equal final outcomes:

- Sender started with \$5, spent \$0.98, has \$4.02 + return
- I started with \$0, got \$2.94, keep \$2.94 - return
- For equal outcomes:  $\$4.02 + \text{return} = \$2.94 - \text{return}$
- $2 * \text{return} = \$2.94 - \$4.02 = -\$1.08$
- This gives negative, which doesn't make sense

Let me think about reciprocity:

- The sender trusted me with \$0.98
- This created value of \$1.96 (the difference between 3x multiplier and original)
- A reasonable return might be to give back enough that we both benefit from the cooperation
- Testing different return amounts: returning \$1.47 would give us nearly equal final gains (\$0.49 for them, \$1.47 for me), while \$1.50 provides them \$0.52 in gains. Even \$1.20 leaves them with only \$0.22 in gains while I keep \$1.74....

**usage:**

```
{
  "input_tokens": 938,
  "output_tokens": 1003,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$8.1.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my experience so far:

From my earnings progression:

- Round 3: I sent \$2, but my earnings only increased by \$3 (the \$3 I kept), suggesting I got \$0 back
- Round 4: I received \$2.94 and returned \$1.50, showing goodwill

The receivers I've encountered have shown poor reciprocity (0% and 18% return rates, though communication noise may be a factor). However, these might be different people each round.

Given this pattern, I need to be more cautious while still leaving room for potential cooperation. The 3x multiplier means even modest returns can be profitable.

I'll send a smaller amount to limit risk while testing if this receiver will reciprocate:

```
'''json
{"send": 1}
'''
```

This sends \$1, creating \$3 for the receiver. If they return even \$1-1.50, we both benefit. If they don't reciprocate, my loss is limited to \$1.

**Reasoning:**

Let me track my earnings:

- Round 1: I sent \$3, which should have given me back something. My total after all rounds up

to Round 2 was \$3.66. So I kept  $\$5 - \$3 = \$2$ , plus whatever was returned = \$3.66. So I received \$1.66 back from round 1.

- Round 2: I was receiver, sender sent \$0, so no change
- Round 3: I sent (need to calculate). Total after round 3 was still \$3.66? No wait, the prompt says "Your total earnings across all rounds are \$3.66" in Round 3. Let me recalculate.

Actually, let me track this more carefully:

- Round 1: I was sender, sent \$3. Total earnings after rounds 1-2 was \$3.66
- Round 2: I was receiver, got \$0. Total after rounds 1-2 was \$3.66
- Round 3: I was sender, sent \$2. Total after rounds 1-3 was \$6.66
- Round 4: I was receiver, got \$2.94, returned \$1.50. Total after rounds 1-4 was \$8.10

So in Round 3, I went from \$3.66 to \$6.66 = +\$3.00

I sent \$2, so I kept \$3 from my endowment

To gain \$3 total, I needed the return to be \$0

So the receiver in Round 3 returned \$0! That's disappointing.

In Round 4, the sender sent me \$0.98 (became \$2.94). I returned \$1.50. I gained  $\$2.94 - \$1.50 = \$1.44$ , which matches:  $\$6.66 + \$1.44 = \$8.10$  ?

So my experience:

- Round 1: Sent \$3 (became \$9), got back \$1.66 (18% return)
- Round 3: Sent \$2 (became \$6), got back \$0 (0% return)
- Round 4: Received \$0.98 (became \$2.94), returned \$1.50

The reciprocity has been weak overall. Still, the 3x multiplier means even modest returns create value. I'm torn between sending nothing to protect myself and sending a small amount to test whether cooperation might work. If I send \$0, I get nothing back. A small send could still be worthwhile if receivers return even 20-30%....

```
Usage: {  
  "input_tokens": 1195,  
  "output_tokens": 992,  
  "reasoning_tokens": 0  
}
```

#### GAME - Agent\_2 - Receiver - dyad\_1

Timestamp: 2026-08-12T09:53:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

#### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### Message 2 role=user

##### content:

Round 1: You are the RECEIVER. The sender sent \$2.05, that is 41.0% of their total endowment. You received \$6.149999999999995. How much do you return to The sender? (0-6.149999999999995)

#### Message 3 role=assistant

##### content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.05 (41% of their endowment)
- I received \$6.15 after the 3x multiplier
- The sender currently has \$2.95 remaining

I want to reward their trust while keeping some benefit from being the receiver. I'll return a reasonable proportion that ensures we both profit.

Returning \$2.50 would give:

- Sender:  $\$2.95 + \$2.50 = \$5.45$  (gain of \$0.45)
- Me:  $\$6.15 - \$2.50 = \$3.65$  (gain of \$3.65)

Both of us benefit, and this return (~41% of what I received) mirrors the sender's level of trust.

{'return': 2.50}

##### reasoning:

Let me analyze this situation:

1. The sender sent \$2.05 out of \$5 (41% of their endowment)
2. This was multiplied by 3x, so I received \$6.15 (approximately)
3. I need to decide how much to return

Key considerations:

- The sender showed moderate trust by sending 41% of their endowment
- If I return nothing, the sender ends up with  $\$5 - \$2.05 = \$2.95$
- If I return a fair amount, we can both benefit
- A common fair strategy is to return enough so both players end up with similar amounts or so the sender at least doesn't lose money

Let me think about fair outcomes:

- Sender currently has:  $\$5 - \$2.05 = \$2.95$
- I currently have:  $\$6.15$
- Total money in system:  $\$2.95 + \$6.15 = \$9.10$

If we split equally: each gets  $\$4.55$

- To achieve this, I'd return:  $\$6.15 - \$4.55 = \$1.60$
- Sender would have:  $\$2.95 + \$1.60 = \$4.55$  ?

Or I could ensure the sender doesn't lose money:

- Sender started with  $\$5$
- Sender sent  $\$2.05$ , has  $\$2.95$  left
- To get back to  $\$5$ , sender needs:  $\$5 - \$2.95 = \$2.05$
- So I could return  $\$2.05$

Or I could return a proportion that seems fair:

- I received  $\$6.15$
- Returning about  $1/3$  to  $1/2$  seems reasonable
- $1/3$  of  $\$6.15 = \$2.05$
- $1/2$  of  $\$6.15 = \$3.075$

Given that the sender sent 41% (a moderate amount showing some trust), I should return a reasonable amount.

Returning  $\$2.50$  would let both of us profit while rewarding their trust. I'll go with  $\$2.50$ , which is roughly 40% of what I received. This gives the sender  $\$5.45$  (a  $\$0.45$  gain) and leaves me with  $\$3.65$  (a  $\$3.65$  gain). The sender took a real risk by sending that much, so returning a meaningful portion acknowledges their willingness to trust....

usage:

```
{
  "input_tokens": 404,
  "output_tokens": 874,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are  $\$3.6499999999999999$ .

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at Round 1, the sender showed moderate trust by sending 41% of their endowment, and I reciprocated with a fair return. This established a cooperative pattern.

Now as the sender, I'll maintain this level of trust and cooperation. Sending a moderate amount signals trust while managing risk.

I'll send  $\$2.00$  (40% of my endowment):

- The receiver will get  $\$6.00$  after the multiplier
- If they reciprocate similarly to how I did in Round 1, both of us should benefit
- This maintains the cooperative dynamic established in Round 1

```
{'send': 2.00}
```

reasoning:

Okay, now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was the receiver:

- Sender sent me  $\$2.05$  (41% of their endowment)
- I received  $\$6.15$
- I returned  $\$2.50$
- So the sender ended up with  $\$2.95 + \$2.50 = \$5.45$
- I ended up with  $\$6.15 - \$2.50 = \$3.65$

The sender in Round 1 showed moderate trust by sending 41% of their endowment, and I reciprocated with a return that was about 41% of what I received. This established a pattern of moderate cooperation.

Now I'm the sender. I need to decide how much to send.

Key considerations:

1. The previous sender sent 41% and got a decent return
2. I should signal trust but also be strategic
3. There's communication noise mentioned, so amounts may differ from what's actually sent
4. I want to maximize my total earnings

If I send \$2 (40%):

- Receiver gets \$6
- If they return similarly to how I did (~41%), they might return \$2.50
- I'd have:  $\$3 + \$2.50 = \$5.50$

If I send \$2.50 (50%):

- Receiver gets \$7.50
- If they return ~40%, they might return \$3
- I'd have:  $\$2.50 + \$3 = \$5.50$

If I send \$3 (60%):

- Receiver gets \$9
- If they return ~40%, they might return \$3.60
- I'd have:  $\$2 + \$3.60 = \$5.60$

If I send \$1.50 (30%):

...

**usage:**

```
{
  "input_tokens": 644,
  "output_tokens": 691,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.6499999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 7 role=assistant**

**content:**

The sender sent \$0 this round, so I received \$0. Since I have nothing to return, my only option is:

```
{'return': 0}
```

**reasoning:**

The sender sent me \$0 this round, so I received \$0 after multiplication.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust me
2. They're being strategic
3. There might be communication noise mentioned in the instructions

Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

This is straightforward - there's nothing to return.

**usage:**

```
{
  "input_tokens": 851,
  "output_tokens": 148,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.6499999999999995.

How much do you send to the receiver this round? (0-5)

**Message 9 role=assistant**

**content:**

Looking at the pattern so far:

- Round 1: I cooperated well as receiver, returning \$2.50
- Round 2: I sent \$2.00 but apparently received nothing back (my earnings suggest \$0 return)
- Round 3: Sender sent me \$0

There's been a breakdown in cooperation, though the note about communication noise means perceptions might differ from reality.

I'll send a moderate amount to signal willingness to cooperate while being cautious given recent experiences. Sending \$1.50 allows me to:

- Test if cooperation can be rebuilt
- Limit potential losses if trust isn't reciprocated
- Keep open the possibility of mutual benefit

```
{'send': 1.50}
```

**reasoning:**

Let me review what's happened:

Round 1: I was receiver, got \$6.15, returned \$2.50

Round 2: I was sender, sent \$2.00

Round 3: I was receiver, sender sent \$0, I returned \$0

My total earnings are \$6.65.

Round 3 is concerning - the sender sent me \$0. This could mean:

1. They didn't trust me to return anything
2. They're being cautious or strategic
3. There might be communication noise affecting their perception
4. They're testing cooperation levels

The instructions mentioned there may be communication noise - amounts might differ from what was actually sent/returned.

In Round 2, I sent \$2.00. I don't know what they returned yet (that's not shown in my earnings update for round 3).

Wait, let me recalculate my earnings:

- Round 1: Started with \$0, received \$6.15, returned \$2.50, kept \$3.65
- Round 2: Started with \$5, sent \$2.00, kept \$3.00 initially. Need to add what was returned to me.
- Round 3: Received \$0, returned \$0, gained \$0

Total earnings = \$6.65

From Round 1: \$3.65

From Round 3: \$0

So from Round 2: \$6.65 - \$3.65 - \$0 = \$3.00

This means in Round 2, I sent \$2.00 and got back \$0. The receiver didn't return anything!

So the pattern is:

- Round 1: Sender sent me \$2.05, I returned \$2.50 (good cooperation from me)
- Round 2: I sent \$2.00, receiver returned \$0 (betrayal!)
- Round 3: Sender sent me \$0 (maybe retaliating? or cautious?)

The zero in Round 3 could be retaliation for my Round 2 behavior, or it might stem from communication distortion about what I actually returned in Round 1. Since each round likely involves different partners, the person who withdrew...

**usage:**

```
{
  "input_tokens": 937,
  "output_tokens": 851,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.36.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

The sender sent \$0 this round, so I received \$0. With nothing received, I have nothing to return:

```
{'return': 0}
```

**Reasoning:**

The sender sent \$0 this round, so I received \$0. I have nothing to return.

This is straightforward - I can only return 0 since I received nothing.

Usage: {

"input\_tokens": 1158,

"output\_tokens": 82,

"reasoning\_tokens": 0

}

=====

**ROUND 6**

=====

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

**Pairings:**

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_2 (Briar) sent \$1; received \$3; Agent\_1 (Aster) returned \$0.6; payoffs Agent\_2 (Briar)=\$4.6, Agent\_1 (Aster)=\$2.4; noise/visible: sent visible=\$0.36, received visible=\$1.08, returned visible=\$0.39

Action validation: {"returned": {"amount": 0.6, "clamped": false, "max": 3.0, "min": 0, "raw": 0.6}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

"Agent\_1": 22.9,

"Agent\_2": 28.1

}

Visible balances after round: {

"Agent\_1": 13.39,

"Agent\_2": 15.75

}

**LLM CALLS**

**GAME - Agent\_2 - Sender - dyad\_1**

Timestamp: 2026-08-12T09:53:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

"dyad\_id": "dyad\_1",

"move\_index": 0,

"opponent\_id": "Agent\_1",

"pairing": {

"agent\_names": {

"Agent\_1": "Aster",

"Agent\_2": "Briar"

},

"agents": [

"Agent\_2",

"Agent\_1"

],

"investor": "Agent\_2",

"roles": {

"Agent\_1": "trustee",

"Agent\_2": "investor"

},

"trustee": "Agent\_1"

},

"role": "investor",

"role\_label": "Sender",

"round": 6,

"task": "game",

```
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$2.05, that is 41.0% of their total endowment. You received \$6.149999999999995. How much do you return to The sender? (0-6.149999999999995)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed moderate trust by sending \$2.05 (41% of their endowment)
- I received \$6.15 after the 3x multiplier
- The sender currently has \$2.95 remaining

I want to reward their trust while keeping some benefit from being the receiver. I'll return a reasonable proportion that ensures we both profit.

Returning \$2.50 would give:

- Sender: \$2.95 + \$2.50 = \$5.45 (gain of \$0.45)
- Me: \$6.15 - \$2.50 = \$3.65 (gain of \$3.65)

Both of us benefit, and this return (~41% of what I received) mirrors the sender's level of trust.

{'return': 2.50}

**reasoning:**

Let me analyze this situation:

1. The sender sent \$2.05 out of \$5 (41% of their endowment)
2. This was multiplied by 3x, so I received \$6.15 (approximately)
3. I need to decide how much to return



Key considerations:

- The sender showed moderate trust by sending 41% of their endowment
- If I return nothing, the sender ends up with  $\$5 - \$2.05 = \$2.95$
- If I return a fair amount, we can both benefit
- A common fair strategy is to return enough so both players end up with similar amounts or so the sender at least doesn't lose money

Let me think about fair outcomes:

- Sender currently has:  $\$5 - \$2.05 = \$2.95$
- I currently have:  $\$6.15$
- Total money in system:  $\$2.95 + \$6.15 = \$9.10$

If we split equally: each gets  $\$4.55$

- To achieve this, I'd return:  $\$6.15 - \$4.55 = \$1.60$
- Sender would have:  $\$2.95 + \$1.60 = \$4.55$  ?

Or I could ensure the sender doesn't lose money:

- Sender started with  $\$5$
- Sender sent  $\$2.05$ , has  $\$2.95$  left
- To get back to  $\$5$ , sender needs:  $\$5 - \$2.95 = \$2.05$
- So I could return  $\$2.05$

Or I could return a proportion that seems fair:

- I received  $\$6.15$
- Returning about  $1/3$  to  $1/2$  seems reasonable
- $1/3$  of  $\$6.15 = \$2.05$
- $1/2$  of  $\$6.15 = \$3.075$

Given that the sender sent 41% (a moderate amount showing some trust), I should return a reasonable amount.

Returning  $\$2.50$  would let both of us profit while rewarding their trust. I'll go with  $\$2.50$ , which is roughly 40% of what I received. This gives the sender  $\$5.45$  (a  $\$0.45$  gain) and leaves me with  $\$3.65$  (a  $\$3.65$  gain). The sender took a real risk by sending that much, so returning a meaningful portion acknowledges their willingness to trust....

usage:

```
{
  "input_tokens": 404,
  "output_tokens": 874,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are  $\$3.6499999999999999$ .

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at Round 1, the sender showed moderate trust by sending 41% of their endowment, and I reciprocated with a fair return. This established a cooperative pattern.

Now as the sender, I'll maintain this level of trust and cooperation. Sending a moderate amount signals trust while managing risk.

I'll send  $\$2.00$  (40% of my endowment):

- The receiver will get  $\$6.00$  after the multiplier
- If they reciprocate similarly to how I did in Round 1, both of us should benefit
- This maintains the cooperative dynamic established in Round 1

```
{'send': 2.00}
```

reasoning:

Okay, now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was the receiver:

- Sender sent me  $\$2.05$  (41% of their endowment)
- I received  $\$6.15$
- I returned  $\$2.50$
- So the sender ended up with  $\$2.95 + \$2.50 = \$5.45$

- I ended up with  $\$6.15 - \$2.50 = \$3.65$

The sender in Round 1 showed moderate trust by sending 41% of their endowment, and I reciprocated with a return that was about 41% of what I received. This established a pattern of moderate cooperation.

Now I'm the sender. I need to decide how much to send.

Key considerations:

1. The previous sender sent 41% and got a decent return
2. I should signal trust but also be strategic
3. There's communication noise mentioned, so amounts may differ from what's actually sent
4. I want to maximize my total earnings

If I send \$2 (40%):

- Receiver gets \$6
- If they return similarly to how I did (~41%), they might return \$2.50
- I'd have:  $\$3 + \$2.50 = \$5.50$

If I send \$2.50 (50%):

- Receiver gets \$7.50
- If they return ~40%, they might return \$3
- I'd have:  $\$2.50 + \$3 = \$5.50$

If I send \$3 (60%):

- Receiver gets \$9
- If they return ~40%, they might return \$3.60
- I'd have:  $\$2 + \$3.60 = \$5.60$

If I send \$1.50 (30%):

...

**usage:**

```
{
  "input_tokens": 644,
  "output_tokens": 691,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.6499999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 7 role=assistant**

**content:**

The sender sent \$0 this round, so I received \$0. Since I have nothing to return, my only option is:

```
{'return': 0}
```

**reasoning:**

The sender sent me \$0 this round, so I received \$0 after multiplication.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust me
2. They're being strategic
3. There might be communication noise mentioned in the instructions

Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

This is straightforward - there's nothing to return.

**usage:**

```
{
  "input_tokens": 851,
  "output_tokens": 148,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.6499999999999995.

How much do you send to the receiver this round? (0-5)

**Message 9 role=assistant**

**content:**

Looking at the pattern so far:

- Round 1: I cooperated well as receiver, returning \$2.50
- Round 2: I sent \$2.00 but apparently received nothing back (my earnings suggest \$0 return)
- Round 3: Sender sent me \$0

There's been a breakdown in cooperation, though the note about communication noise means perceptions might differ from reality.

I'll send a moderate amount to signal willingness to cooperate while being cautious given recent experiences. Sending \$1.50 allows me to:

- Test if cooperation can be rebuilt
- Limit potential losses if trust isn't reciprocated
- Keep open the possibility of mutual benefit

{'send': 1.50}

**reasoning:**

Let me review what's happened:

Round 1: I was receiver, got \$6.15, returned \$2.50

Round 2: I was sender, sent \$2.00

Round 3: I was receiver, sender sent \$0, I returned \$0

My total earnings are \$6.65.

Round 3 is concerning - the sender sent me \$0. This could mean:

1. They didn't trust me to return anything
2. They're being cautious or strategic
3. There might be communication noise affecting their perception
4. They're testing cooperation levels

The instructions mentioned there may be communication noise - amounts might differ from what was actually sent/returned.

In Round 2, I sent \$2.00. I don't know what they returned yet (that's not shown in my earnings update for round 3).

Wait, let me recalculate my earnings:

- Round 1: Started with \$0, received \$6.15, returned \$2.50, kept \$3.65
- Round 2: Started with \$5, sent \$2.00, kept \$3.00 initially. Need to add what was returned to me.
- Round 3: Received \$0, returned \$0, gained \$0

Total earnings = \$6.65

From Round 1: \$3.65

From Round 3: \$0

So from Round 2:  $\$6.65 - \$3.65 - \$0 = \$3.00$

This means in Round 2, I sent \$2.00 and got back \$0. The receiver didn't return anything!

So the pattern is:

- Round 1: Sender sent me \$2.05, I returned \$2.50 (good cooperation from me)
- Round 2: I sent \$2.00, receiver returned \$0 (betrayal!)
- Round 3: Sender sent me \$0 (maybe retaliating? or cautious?)

The zero in Round 3 could be retaliation for my Round 2 behavior, or it might stem from communication distortion about what I actually returned in Round 1. Since each round likely involves different partners, the person who withdrew...

**usage:**

```
{
  "input_tokens": 937,
  "output_tokens": 851,
```

```

    "reasoning_tokens": 0
  }
Message 10 role=user
  content:
    Round 5

    You are the RECEIVER this round. Your total earnings across all rounds are $11.36.
    This round, The sender sent you $0, so you received $0. How much do you return to The sender? (0-0)
Message 11 role=assistant
  content:
    The sender sent $0 this round, so I received $0. With nothing received, I have nothing to return:

    {'return': 0}
  reasoning:
    The sender sent $0 this round, so I received $0. I have nothing to return.

    This is straightforward - I can only return 0 since I received nothing.
  usage:
    {
      "input_tokens": 1158,
      "output_tokens": 82,
      "reasoning_tokens": 0
    }
Message 12 role=user
  content:
    Round 6

    You are the SENDER this round. Your total earnings across all rounds are $11.36.
    How much do you send to the receiver this round? (0-5)
Assistant output:
  Looking at the pattern:

  - Round 1: Good cooperation when I was receiver
  - Round 2: I sent $2.00, got little/nothing back
  - Round 3: Received nothing as receiver
  - Round 4: I sent $1.50, got some return (earnings increased to $11.36)
  - Round 5: Received nothing as receiver again

  Cooperation has been inconsistent, though the communication noise mentioned could be affecting
  perceptions on both sides. The receiver in Round 4 did return something when I sent, which
  is a positive signal.

  I'll send a small amount to signal continued willingness to cooperate while being cautious:

  {'send': 1.00}
Reasoning:
  Let me analyze the pattern of play:

  Round 1: I was RECEIVER
  - Sender sent $2.05 (41%)
  - I received $6.15
  - I returned $2.50
  - My earnings: $3.65

  Round 2: I was SENDER
  - I sent $2.00
  - My total earnings after Round 2: $6.65
  - This means I gained $3.00 in Round 2 ( $6.65 - 3.65 = 3.00$ )
  - I started with $5, sent $2, left with $3
  - If I got back nothing, I'd have $3 total, but I have $6.65 total
  - Actually wait, let me recalculate. After Round 1, I had $3.65
  - In Round 2 as sender, I get $5 new endowment, send $2, keep $3
  - So I should have  $3.65 + \$3 = \$6.65$  if I got nothing back
  - My total is $6.65, so I got $0 back in Round 2

  Round 3: I was RECEIVER
  - Sender sent $0
  - I received $0

```

- I returned \$0
- Total earnings stayed at \$6.65

Round 4: I was SENDER

- I sent \$1.50
- My total earnings after Round 4: \$6.65
- Wait, that doesn't make sense. Let me check again.
- Before Round 4, I had \$6.65
- In Round 4, I get new \$5 endowment, send \$1.50, keep \$3.50
- If I got nothing back, I'd have  $\$6.65 + \$3.50 = \$10.15$
- But my total after Round 4 should be reflected in what's stated before Round 5
- The prompt shows \$11.36 total before Round 5, so after Round 4 I had \$11.36
- That's a gain of \$4.71 from the \$6.65 I started with
- Since I kept \$3.50 from my endowment, the receiver must have returned \$1.21

Round 5: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My total earnings stayed at \$11.36

Now for Round 6: I'm SENDER again

- My total is still \$11.36

Looking at the pattern, cooperation has been sporadic. Round 1 showed genuine reciprocity when the sender trusted me with 41% and I returned generously. But Round 2 went nowhere-I sent 40% and got nothing back. Round 3 and 5 both had senders sending nothing. Round 4 was the only other time I sent (30%), and while I did get \$1.21 back, that's only 27% reciprocation on what they received-quite weak. The communication issues mentioned might be playing a role in this inconsistency....

```
Usage: {
  "input_tokens": 1237,
  "output_tokens": 1167,
  "reasoning_tokens": 0
}
```

#### GAME - Agent\_1 - Receiver - dyad\_1

Timestamp: 2026-08-12T09:53:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

```
Message 1 role=system
content:
```

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to decide how much trust to place in an unknown receiver.

Since this appears to be a multi-round game, establishing cooperation could be beneficial long-term. Sending money creates value (3x multiplier), but I need the receiver to share the gains fairly.

I'll send a moderate amount that signals trust while managing risk:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return a fair share (around \$4-5), we both benefit compared to not cooperating. This also lets me gauge their trustworthiness for future rounds.

**reasoning:**

This is the first round of a trust game where I'm playing as the SENDER. Let me think through the strategic considerations:

1. I start with \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned

Key considerations:

- If I send nothing, I keep \$5 and the receiver gets nothing
- If I send everything (\$5), the receiver gets \$15, and then they decide what to return
- There's a note about "communication noise" - amounts may differ slightly from what was actually sent/returned

This is a classic trust game. The optimal strategy depends on:

1. Whether I trust the receiver to reciprocate
2. Whether there will be future rounds with the same person (the prompt mentions "multiple rounds")
3. The communication noise factor

For the first round, I don't have any information about the receiver's behavior. Common approaches:

- Send a moderate amount to signal trust and willingness to cooperate
- See how the receiver responds to gauge their trustworthiness
- In repeated games, cooperation often emerges

A reasonable first move would be to send around \$2-3 to:

- Show willingness to trust and cooperate
- Not risk everything on an unknown partner
- Leave room to adjust strategy based on their response

I'll send \$3 as a fairly trusting opening move that still hedges some risk.

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$3.66.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

Since the sender sent me \$0, I received \$0 (even after the 3x multiplier).

With \$0 received, I have nothing to return. The only possible amount is:

```
'''json
{"return": 0}
'''
```

**reasoning:**

Let me understand the situation:

Round 1: I was SENDER and sent \$3

- I kept:  $\$5 - \$3 = \$2$
- Receiver got:  $\$3 \times 3 = \$9$
- My total earnings are \$3.66

This means I must have received \$1.66 back from the receiver in Round 1.

So:  $\$2$  (kept) +  $\$1.66$  (returned) =  $\$3.66$  ?

Round 2: I am RECEIVER

- The sender sent me \$0
- I received \$0 (since  $\$0 \times 3 = \$0$ )
- I can only return between \$0 and \$0

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

**usage:**

```
{
  "input_tokens": 567,
  "output_tokens": 245,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$3.66.

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at my situation:

From Round 1, I can calculate that the receiver only returned \$1.66 out of the \$9 they received (about 18% return rate), which shows low reciprocity. However, I should note the warning about communication noise - the actual amounts may have differed.

Round 2 didn't affect my earnings since the sender sent \$0.

For Round 3, I'm facing uncertainty:

- This could be a different receiver
- Communication noise may have affected Round 1 observations
- I need to balance potential gains against the risk of poor reciprocity

I'll send a smaller amount to test this receiver's trustworthiness while limiting my downside risk:

```
'''json
{"send": 2}
'''
```

This sends \$2, which becomes \$6 for the receiver. It's enough to create mutual benefit if they reciprocate fairly, but limits my loss if they don't.

**reasoning:**

Let me analyze the situation:

Round 1: I was the SENDER and sent \$3

- My total earnings are now \$3.66
- If I sent \$3, I kept \$5 - \$3 = \$2
- The receiver got  $3 \times 3 = 9$
- They returned some amount to me
- So I have:  $2 + \text{returned amount} = 3.66$
- This means they returned \$1.66 to me out of the \$9 they received

That's a pretty poor return rate ( $1.66/9 \approx 18.4\%$ ). They kept \$7.34 out of \$9.

Round 2: I was the RECEIVER

- The sender sent me \$0
- So I received \$0 and returned \$0
- This didn't affect my earnings

Now in Round 3, I'm the SENDER again.

The receiver from Round 1 showed low trustworthiness by only returning \$1.66 out of \$9. However, I should note that the instructions mention "communication noise" - the amounts I see may differ from what was actually sent/returned.

Given that:

1. The previous receiver showed poor reciprocity (only returned ~18% of what they received)
2. There may be communication noise affecting what I observe
3. This might be a different receiver (it's a multi-round game, roles change)

I have a few options:

- Send \$0 to protect my earnings (risk-averse)
- Send a small amount to test this receiver
- Send a moderate amount hoping for better reciprocity

Given the poor experience in Round 1, I'll be more cautious. I'll send a smaller amount to test this receiver's trustworthiness:

I'll send \$2 this time.

**usage:**

```
{
  "input_tokens": 668,
  "output_tokens": 617,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**



**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$6.66.

This round, The sender sent you \$0.98, so you received \$2.94. How much do you return to The sender? (0-2.94)

**Message 9 role=assistant****content:**

Looking at this situation:

The sender sent me \$0.98, which became \$2.94 after the 3x multiplier. This created value through cooperation.

To encourage continued cooperation and show trustworthiness, I should return a fair amount. Let me consider the gains:

- Total value created from cooperation: \$1.96 (\$2.94 - \$0.98)
- A fair return would split these gains reasonably

If I return \$1.50:

- Sender gains: \$5 - \$0.98 + \$1.50 = \$5.52 (net gain: \$0.52)
- I gain: \$2.94 - \$1.50 = \$1.44 (net gain: \$1.44)

This gives us both positive gains from cooperating, with me gaining slightly more since I received the multiplied amount.

```
'''json
{"return": 1.50}
'''
```

**reasoning:**

Let me analyze this situation:

The sender sent me \$0.98, which was multiplied by 3x to give me \$2.94.

I need to decide how much to return. Let me think about fairness and cooperation:

1. The sender gave up \$0.98 from their \$5 endowment
2. This created \$2.94 for me
3. The total value created is \$2.94 - \$0.98 = \$1.96 in additional value due to the multiplier

A fair split might consider:

- Equal split of gains: The sender sacrificed \$0.98, I got \$2.94. If I return nothing, sender has \$4.02 and I have \$2.94. If we split the \$1.96 gain equally, I should return enough so we each gain about \$0.98.
- To give the sender a total gain of ~\$0.98, they need: \$5 - \$0.98 + return = \$5.98, so return = \$0.98
- That seems too low. Let me think differently.

Alternative approach - equal final outcomes:

- Sender started with \$5, spent \$0.98, has \$4.02 + return
- I started with \$0, got \$2.94, keep \$2.94 - return
- For equal outcomes: \$4.02 + return = \$2.94 - return
- 2\*return = \$2.94 - \$4.02 = -\$1.08
- This gives negative, which doesn't make sense

Let me think about reciprocity:

- The sender trusted me with \$0.98
- This created value of \$1.96 (the difference between 3x multiplier and original)
- A reasonable return might be to give back enough that we both benefit from the cooperation
- Testing different return amounts: returning \$1.47 would give us nearly equal final gains (\$0.49 for them, \$1.47 for me), while \$1.50 provides them \$0.52 in gains. Even \$1.20 leaves them with only \$0.22 in gains while I keep \$1.74....

**usage:**

```
{
  "input_tokens": 938,
  "output_tokens": 1003,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$8.1.

How much do you send to the receiver this round? (0-5)

**Message 11 role=assistant**

**content:**

Looking at my experience so far:

From my earnings progression:

- Round 3: I sent \$2, but my earnings only increased by \$3 (the \$3 I kept), suggesting I got \$0 back
- Round 4: I received \$2.94 and returned \$1.50, showing goodwill

The receivers I've encountered have shown poor reciprocity (0% and 18% return rates, though communication noise may be a factor). However, these might be different people each round.

Given this pattern, I need to be more cautious while still leaving room for potential cooperation. The 3x multiplier means even modest returns can be profitable.

I'll send a smaller amount to limit risk while testing if this receiver will reciprocate:

```
'''json
{"send": 1}
'''
```

This sends \$1, creating \$3 for the receiver. If they return even \$1-1.50, we both benefit. If they don't reciprocate, my loss is limited to \$1.

**reasoning:**

Let me track my earnings:

- Round 1: I sent \$3, which should have given me back something. My total after all rounds up to Round 2 was \$3.66. So I kept \$5 - \$3 = \$2, plus whatever was returned = \$3.66. So I received \$1.66 back from round 1.
- Round 2: I was receiver, sender sent \$0, so no change
- Round 3: I sent (need to calculate). Total after round 3 was still \$3.66? No wait, the prompt says "Your total earnings across all rounds are \$3.66" in Round 3. Let me recalculate.

Actually, let me track this more carefully:

- Round 1: I was sender, sent \$3. Total earnings after rounds 1-2 was \$3.66
- Round 2: I was receiver, got \$0. Total after rounds 1-2 was \$3.66
- Round 3: I was sender, sent \$2. Total after rounds 1-3 was \$6.66
- Round 4: I was receiver, got \$2.94, returned \$1.50. Total after rounds 1-4 was \$8.10

So in Round 3, I went from \$3.66 to \$6.66 = +\$3.00

I sent \$2, so I kept \$3 from my endowment

To gain \$3 total, I needed the return to be \$0

So the receiver in Round 3 returned \$0! That's disappointing.

In Round 4, the sender sent me \$0.98 (became \$2.94). I returned \$1.50. I gained \$2.94 - \$1.50 = \$1.44, which matches: \$6.66 + \$1.44 = \$8.10 ?

So my experience:

- Round 1: Sent \$3 (became \$9), got back \$1.66 (18% return)
- Round 3: Sent \$2 (became \$6), got back \$0 (0% return)
- Round 4: Received \$0.98 (became \$2.94), returned \$1.50

The reciprocity has been weak overall. Still, the 3x multiplier means even modest returns create value. I'm torn between sending nothing to protect myself and sending a small amount to test whether cooperation might work. If I send \$0, I get nothing back. A small send could still be worthwhile if receivers return even 20-30%...

**usage:**

```
{
  "input_tokens": 1195,
  "output_tokens": 992,
  "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.91.

This round, The sender sent you \$0.36, so you received \$1.08. How much do you return to The sender? (0-1.08)

**Assistant output:**

Looking at this situation:

The sender sent me \$0.36, which became \$1.08 after the 3x multiplier. This is a very small amount, suggesting they have low trust - likely from poor past experiences.

To demonstrate trustworthiness and potentially build cooperation:

Total value created from cooperation:  $\$1.08 - \$0.36 = \$0.72$

If I return \$0.60:

- Sender gains:  $\$5 - \$0.36 + \$0.60 = \$5.24$  (net +\$0.24)
- I gain:  $\$1.08 - \$0.60 = \$0.48$  (net +\$0.48)

This shows strong reciprocity (~56% return rate) which could encourage more trust and larger exchanges in future rounds, benefiting both of us.

```
'''json
{"return": 0.60}
'''
```

**Reasoning:**

Let me analyze this situation:

The sender sent me \$0.36, which became \$1.08 after the 3x multiplier.

This is a very small amount - they're being quite cautious, likely due to past experiences with low reciprocity.

Let me think about what's fair:

- Total value created:  $\$1.08 - \$0.36 = \$0.72$
- A fair split of gains might be around half each

If I return \$0.54:

- Sender:  $\$5 - \$0.36 + \$0.54 = \$5.18$  (gain of \$0.18)
- Me:  $\$1.08 - \$0.54 = \$0.54$  (gain of \$0.54)

This gives us both positive gains, with me gaining more since I received the multiplied amount.

Alternatively, if I return \$0.70:

- Sender:  $\$5 - \$0.36 + \$0.70 = \$5.34$  (gain of \$0.34)
- Me:  $\$1.08 - \$0.70 = \$0.38$  (gain of \$0.38)

This is closer to an equal split of the total value created.

Given that:

1. The sender sent a very small amount (showing low trust)
2. I want to demonstrate trustworthiness to encourage future cooperation
3. Even a modest return shows good faith
4. The amounts are small, so being generous could pay off

I'll return about 50-65% of what I received, which shows strong reciprocity and might encourage the sender to trust more in future rounds.

Let me return \$0.60:

...

```
Usage: {
  "input_tokens": 1485,
  "output_tokens": 669,
  "reasoning_tokens": 0
}
```

=====

**ROUND 7**

=====

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

**Pairings:**

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_1 (Aster) sent \$2; received \$6; Agent\_2 (Briar) returned \$0; payoffs Agent\_1 (Aster)=\$3, Agent\_2 (Briar)=\$6; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.85

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent\_1": 25.9,

"Agent\_2": 34.1

}

Visible balances after round: {

"Agent\_1": 17.240000000000002,

"Agent\_2": 15.75

}

**LLM CALLS**

**GAME - Agent\_1 - Sender - dyad\_1**

Timestamp: 2026-08-12T09:53:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

"dyad\_id": "dyad\_1",

"move\_index": 0,

"opponent\_id": "Agent\_2",

"pairing": {

"agent\_names": {

"Agent\_1": "Aster",

"Agent\_2": "Briar"

},

"agents": [

"Agent\_1",

"Agent\_2"

],

"investor": "Agent\_1",

"roles": {

"Agent\_1": "investor",

"Agent\_2": "trustee"

},

"trustee": "Agent\_2"

},

"role": "investor",

"role\_label": "Sender",

"round": 7,

"task": "game",

"task\_index": 0

}

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### Message 2 role=user

##### content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

#### Message 3 role=assistant

##### content:

Looking at this first round, I need to decide how much trust to place in an unknown receiver.

Since this appears to be a multi-round game, establishing cooperation could be beneficial long-term. Sending money creates value (3x multiplier), but I need the receiver to share the gains fairly.

I'll send a moderate amount that signals trust while managing risk:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return a fair share (around \$4-5), we both benefit compared to not cooperating. This also lets me gauge their trustworthiness for future rounds.

##### reasoning:

This is the first round of a trust game where I'm playing as the SENDER. Let me think through the strategic considerations:

1. I start with \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned

##### Key considerations:

- If I send nothing, I keep \$5 and the receiver gets nothing
- If I send everything (\$5), the receiver gets \$15, and then they decide what to return
- There's a note about "communication noise" - amounts may differ slightly from what was actually sent/returned

This is a classic trust game. The optimal strategy depends on:

1. Whether I trust the receiver to reciprocate
2. Whether there will be future rounds with the same person (the prompt mentions "multiple rounds")
3. The communication noise factor

For the first round, I don't have any information about the receiver's behavior. Common approaches:

- Send a moderate amount to signal trust and willingness to cooperate
- See how the receiver responds to gauge their trustworthiness
- In repeated games, cooperation often emerges

A reasonable first move would be to send around \$2-3 to:

- Show willingness to trust and cooperate
- Not risk everything on an unknown partner
- Leave room to adjust strategy based on their response

I'll send \$3 as a fairly trusting opening move that still hedges some risk.

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$3.66.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

Since the sender sent me \$0, I received \$0 (even after the 3x multiplier).

With \$0 received, I have nothing to return. The only possible amount is:

```
'''json
{"return": 0}
'''
```

**reasoning:**

Let me understand the situation:

Round 1: I was SENDER and sent \$3

- I kept:  $\$5 - \$3 = \$2$
- Receiver got:  $\$3 \times 3 = \$9$
- My total earnings are \$3.66

This means I must have received \$1.66 back from the receiver in Round 1.

So:  $\$2$  (kept) +  $\$1.66$  (returned) =  $\$3.66$  ?

Round 2: I am RECEIVER

- The sender sent me \$0
- I received \$0 (since  $\$0 \times 3 = \$0$ )
- I can only return between \$0 and \$0

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

**usage:**

```
{
  "input_tokens": 567,
  "output_tokens": 245,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$3.66.

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at my situation:

From Round 1, I can calculate that the receiver only returned \$1.66 out of the \$9 they received (about 18% return rate), which shows low reciprocity. However, I should note the warning about communication noise - the actual amounts may have differed.

Round 2 didn't affect my earnings since the sender sent \$0.

For Round 3, I'm facing uncertainty:

- This could be a different receiver
- Communication noise may have affected Round 1 observations
- I need to balance potential gains against the risk of poor reciprocity

I'll send a smaller amount to test this receiver's trustworthiness while limiting my downs

ide risk:

```
'''json
{"send": 2}
'''
```

This sends \$2, which becomes \$6 for the receiver. It's enough to create mutual benefit if they reciprocate fairly, but limits my loss if they don't.

**reasoning:**

Let me analyze the situation:

Round 1: I was the SENDER and sent \$3

- My total earnings are now \$3.66
- If I sent \$3, I kept \$5 - \$3 = \$2
- The receiver got  $\$3 \times 3 = \$9$
- They returned some amount to me
- So I have:  $\$2 + \text{returned amount} = \$3.66$
- This means they returned \$1.66 to me out of the \$9 they received

That's a pretty poor return rate ( $1.66/9 \approx 18.4\%$ ). They kept \$7.34 out of \$9.

Round 2: I was the RECEIVER

- The sender sent me \$0
- So I received \$0 and returned \$0
- This didn't affect my earnings

Now in Round 3, I'm the SENDER again.

The receiver from Round 1 showed low trustworthiness by only returning \$1.66 out of \$9. However, I should note that the instructions mention "communication noise" - the amounts I see may differ from what was actually sent/returned.

Given that:

1. The previous receiver showed poor reciprocity (only returned ~18% of what they received)
2. There may be communication noise affecting what I observe
3. This might be a different receiver (it's a multi-round game, roles change)

I have a few options:

- Send \$0 to protect my earnings (risk-averse)
- Send a small amount to test this receiver
- Send a moderate amount hoping for better reciprocity

Given the poor experience in Round 1, I'll be more cautious. I'll send a smaller amount to test this receiver's trustworthiness:

I'll send \$2 this time.

**usage:**

```
{
  "input_tokens": 668,
  "output_tokens": 617,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$6.66.

This round, The sender sent you \$0.98, so you received \$2.94. How much do you return to the sender? (0-2.94)

**Message 9 role=assistant**

**content:**

Looking at this situation:

The sender sent me \$0.98, which became \$2.94 after the 3x multiplier. This created value through cooperation.

To encourage continued cooperation and show trustworthiness, I should return a fair amount. Let me consider the gains:

- Total value created from cooperation: \$1.96 ( $\$2.94 - \$0.98$ )

- A fair return would split these gains reasonably

If I return \$1.50:

- Sender gains:  $\$5 - \$0.98 + \$1.50 = \$5.52$  (net gain:  $\$0.52$ )
- I gain:  $\$2.94 - \$1.50 = \$1.44$  (net gain:  $\$1.44$ )

This gives us both positive gains from cooperating, with me gaining slightly more since I received the multiplied amount.

```
'''json
{"return": 1.50}
'''
```

**reasoning:**

Let me analyze this situation:

The sender sent me \$0.98, which was multiplied by 3x to give me \$2.94.

I need to decide how much to return. Let me think about fairness and cooperation:

1. The sender gave up \$0.98 from their \$5 endowment
2. This created \$2.94 for me
3. The total value created is  $\$2.94 - \$0.98 = \$1.96$  in additional value due to the multiplier

A fair split might consider:

- Equal split of gains: The sender sacrificed \$0.98, I got \$2.94. If I return nothing, sender has \$4.02 and I have \$2.94. If we split the \$1.96 gain equally, I should return enough so we each gain about \$0.98.
- To give the sender a total gain of ~\$0.98, they need:  $\$5 - \$0.98 + \text{return} = \$5.98$ , so return = \$0.98
- That seems too low. Let me think differently.

Alternative approach - equal final outcomes:

- Sender started with \$5, spent \$0.98, has  $\$4.02 + \text{return}$
- I started with \$0, got \$2.94, keep  $\$2.94 - \text{return}$
- For equal outcomes:  $\$4.02 + \text{return} = \$2.94 - \text{return}$
- $2 * \text{return} = \$2.94 - \$4.02 = -\$1.08$
- This gives negative, which doesn't make sense

Let me think about reciprocity:

- The sender trusted me with \$0.98
- This created value of \$1.96 (the difference between 3x multiplier and original)
- A reasonable return might be to give back enough that we both benefit from the cooperation
- Testing different return amounts: returning \$1.47 would give us nearly equal final gains (\$0.49 for them, \$1.47 for me), while \$1.50 provides them \$0.52 in gains. Even \$1.20 leaves them with only \$0.22 in gains while I keep \$1.74....

**usage:**

```
{
  "input_tokens": 938,
  "output_tokens": 1003,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$8.1.

How much do you send to the receiver this round? (0-5)

**Message 11 role=assistant**

**content:**

Looking at my experience so far:

From my earnings progression:

- Round 3: I sent \$2, but my earnings only increased by \$3 (the \$3 I kept), suggesting I got \$0 back
- Round 4: I received \$2.94 and returned \$1.50, showing goodwill

The receivers I've encountered have shown poor reciprocity (0% and 18% return rates, though communication noise may be a factor). However, these might be different people each round.



und.

Given this pattern, I need to be more cautious while still leaving room for potential cooperation. The 3x multiplier means even modest returns can be profitable.

I'll send a smaller amount to limit risk while testing if this receiver will reciprocate:

```
'''json
{"send": 1}
'''
```

This sends \$1, creating \$3 for the receiver. If they return even \$1-1.50, we both benefit. If they don't reciprocate, my loss is limited to \$1.

**reasoning:**

Let me track my earnings:

- Round 1: I sent \$3, which should have given me back something. My total after all rounds up to Round 2 was \$3.66. So I kept \$5 - \$3 = \$2, plus whatever was returned = \$3.66. So I received \$1.66 back from round 1.
- Round 2: I was receiver, sender sent \$0, so no change
- Round 3: I sent (need to calculate). Total after round 3 was still \$3.66? No wait, the prompt says "Your total earnings across all rounds are \$3.66" in Round 3. Let me recalculate.

Actually, let me track this more carefully:

- Round 1: I was sender, sent \$3. Total earnings after rounds 1-2 was \$3.66
- Round 2: I was receiver, got \$0. Total after rounds 1-2 was \$3.66
- Round 3: I was sender, sent \$2. Total after rounds 1-3 was \$6.66
- Round 4: I was receiver, got \$2.94, returned \$1.50. Total after rounds 1-4 was \$8.10

So in Round 3, I went from \$3.66 to \$6.66 = +\$3.00

I sent \$2, so I kept \$3 from my endowment

To gain \$3 total, I needed the return to be \$0

So the receiver in Round 3 returned \$0! That's disappointing.

In Round 4, the sender sent me \$0.98 (became \$2.94). I returned \$1.50. I gained \$2.94 - \$1.50 = \$1.44, which matches: \$6.66 + \$1.44 = \$8.10 ?

So my experience:

- Round 1: Sent \$3 (became \$9), got back \$1.66 (18% return)
- Round 3: Sent \$2 (became \$6), got back \$0 (0% return)
- Round 4: Received \$0.98 (became \$2.94), returned \$1.50

The reciprocity has been weak overall. Still, the 3x multiplier means even modest returns create value. I'm torn between sending nothing to protect myself and sending a small amount to test whether cooperation might work. If I send \$0, I get nothing back. A small send could still be worthwhile if receivers return even 20-30%...

**usage:**

```
{
  "input_tokens": 1195,
  "output_tokens": 992,
  "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.91.

This round, The sender sent you \$0.36, so you received \$1.08. How much do you return to The sender? (0-1.08)

**Message 13 role=assistant**

**content:**

Looking at this situation:

The sender sent me \$0.36, which became \$1.08 after the 3x multiplier. This is a very small amount, suggesting they have low trust - likely from poor past experiences.

To demonstrate trustworthiness and potentially build cooperation:

Total value created from cooperation: \$1.08 - \$0.36 = \$0.72

If I return \$0.60:  
- Sender gains:  $\$5 - \$0.36 + \$0.60 = \$5.24$  (net +\$0.24)  
- I gain:  $\$1.08 - \$0.60 = \$0.48$  (net +\$0.48)

This shows strong reciprocity (~56% return rate) which could encourage more trust and larger exchanges in future rounds, benefiting both of us.

```
'''json
{"return": 0.60}
'''
```

**reasoning:**

Let me analyze this situation:

The sender sent me \$0.36, which became \$1.08 after the 3x multiplier.

This is a very small amount - they're being quite cautious, likely due to past experiences with low reciprocity.

Let me think about what's fair:

- Total value created:  $\$1.08 - \$0.36 = \$0.72$
- A fair split of gains might be around half each

If I return \$0.54:

- Sender:  $\$5 - \$0.36 + \$0.54 = \$5.18$  (gain of \$0.18)
- Me:  $\$1.08 - \$0.54 = \$0.54$  (gain of \$0.54)

This gives us both positive gains, with me gaining more since I received the multiplied amount.

Alternatively, if I return \$0.70:

- Sender:  $\$5 - \$0.36 + \$0.70 = \$5.34$  (gain of \$0.34)
- Me:  $\$1.08 - \$0.70 = \$0.38$  (gain of \$0.38)

This is closer to an equal split of the total value created.

Given that:

1. The sender sent a very small amount (showing low trust)
2. I want to demonstrate trustworthiness to encourage future cooperation
3. Even a modest return shows good faith
4. The amounts are small, so being generous could pay off

I'll return about 50-65% of what I received, which shows strong reciprocity and might encourage the sender to trust more in future rounds.

Let me return \$0.60:

...

**usage:**

```
{
  "input_tokens": 1485,
  "output_tokens": 669,
  "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.39.  
How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my track record:

As SENDER:

- Round 1: Sent \$3, got back \$1.66 (low reciprocity)
- Round 3: Sent \$2, got back \$0 (no reciprocity)
- Round 5: Sent \$1, got back \$0.81 (best reciprocity rate ~27%)

As RECEIVER, I've been returning 50-56%, demonstrating trustworthiness.

Interestingly, my smallest send (Round 5) got the best return rate. This could be due to:

- Communication noise making small amounts appear larger

- Building reputation from my fair behavior as receiver
- Different receivers with varying trustworthiness

I'll send a moderate amount to test if cooperation is improving:

```
'''json
{"send": 2}
'''
```

This sends \$2 (becomes \$6). If reciprocity continues improving, we both benefit. If not, my loss is limited.

#### Reasoning:

Let me track my earnings:

- Round 1 (SENDER): I sent \$3, ended with \$3.66. So I kept \$2 and got back \$1.66
- Round 2 (RECEIVER): Sender sent \$0, I got \$0. No change to earnings.
- Round 3 (SENDER): Started at \$3.66. I sent \$2. Ended at \$6.66, so gained \$3. This means I kept \$3 and got back \$0.
- Round 4 (RECEIVER): Started at \$6.66. Received \$2.94, returned \$1.50. Gained \$1.44 (\$2.94-\$1.50). Ended at \$8.1.
- Round 5 (SENDER): Started at \$8.1. I sent \$1. Ended at \$12.91, so gained \$4.81. I kept \$4, so I got back \$0.81.
- Round 6 (RECEIVER): Started at \$12.91. Received \$1.08, returned \$0.60. Gained \$0.48 (\$1.08-\$0.60). Ended at \$13.39.
- Round 7 (SENDER): Starting at \$13.39

Looking at patterns when I was SENDER:

- Round 1: Sent \$3, got back \$1.66 (18.4% of the \$9 received)
- Round 3: Sent \$2, got back \$0 (0% of the \$6 received)
- Round 5: Sent \$1, got back \$0.81 (27% of the \$3 received)

Interesting - Round 5 showed better reciprocity (27% return) when I sent less. And I've been showing good reciprocity when I'm a receiver (50% in Round 4, 56% in Round 6).

The pattern suggests:

1. Different receivers each round

...

```
Usage: {
  "input_tokens": 1735,
  "output_tokens": 797,
  "reasoning_tokens": 0
}
```

#### GAME - Agent\_2 - Receiver - dyad\_1

Timestamp: 2026-08-12T09:53:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
```

```
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$2.05, that is 41.0% of their total endowment. You received \$6.1499999999999995. How much do you return to The sender? (0-6.1499999999999995)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed moderate trust by sending \$2.05 (41% of their endowment)
- I received \$6.15 after the 3x multiplier
- The sender currently has \$2.95 remaining

I want to reward their trust while keeping some benefit from being the receiver. I'll return a reasonable proportion that ensures we both profit.

Returning \$2.50 would give:

- Sender: \$2.95 + \$2.50 = \$5.45 (gain of \$0.45)
- Me: \$6.15 - \$2.50 = \$3.65 (gain of \$3.65)

Both of us benefit, and this return (~41% of what I received) mirrors the sender's level of trust.

{'return': 2.50}

**reasoning:**

Let me analyze this situation:

1. The sender sent \$2.05 out of \$5 (41% of their endowment)
2. This was multiplied by 3x, so I received \$6.15 (approximately)
3. I need to decide how much to return

Key considerations:

- The sender showed moderate trust by sending 41% of their endowment
- If I return nothing, the sender ends up with  $\$5 - \$2.05 = \$2.95$
- If I return a fair amount, we can both benefit
- A common fair strategy is to return enough so both players end up with similar amounts or so the sender at least doesn't lose money

Let me think about fair outcomes:

- Sender currently has:  $\$5 - \$2.05 = \$2.95$
- I currently have:  $\$6.15$
- Total money in system:  $\$2.95 + \$6.15 = \$9.10$

If we split equally: each gets  $\$4.55$

- To achieve this, I'd return:  $\$6.15 - \$4.55 = \$1.60$
- Sender would have:  $\$2.95 + \$1.60 = \$4.55$  ?

Or I could ensure the sender doesn't lose money:

- Sender started with  $\$5$
- Sender sent  $\$2.05$ , has  $\$2.95$  left
- To get back to  $\$5$ , sender needs:  $\$5 - \$2.95 = \$2.05$
- So I could return  $\$2.05$

Or I could return a proportion that seems fair:

- I received  $\$6.15$
- Returning about  $1/3$  to  $1/2$  seems reasonable
- $1/3$  of  $\$6.15 = \$2.05$
- $1/2$  of  $\$6.15 = \$3.075$

Given that the sender sent 41% (a moderate amount showing some trust), I should return a reasonable amount.

Returning  $\$2.50$  would let both of us profit while rewarding their trust. I'll go with  $\$2.50$ , which is roughly 40% of what I received. This gives the sender  $\$5.45$  (a  $\$0.45$  gain) and I leave with  $\$3.65$  (a  $\$3.65$  gain). The sender took a real risk by sending that much, so returning a meaningful portion acknowledges their willingness to trust....

usage:

```
{
  "input_tokens": 404,
  "output_tokens": 874,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are  $\$3.6499999999999999$ .

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at Round 1, the sender showed moderate trust by sending 41% of their endowment, and I reciprocated with a fair return. This established a cooperative pattern.

Now as the sender, I'll maintain this level of trust and cooperation. Sending a moderate amount signals trust while managing risk.

I'll send  $\$2.00$  (40% of my endowment):

- The receiver will get  $\$6.00$  after the multiplier
- If they reciprocate similarly to how I did in Round 1, both of us should benefit
- This maintains the cooperative dynamic established in Round 1

```
{'send': 2.00}
```

reasoning:

Okay, now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was the receiver:

- Sender sent me  $\$2.05$  (41% of their endowment)
- I received  $\$6.15$
- I returned  $\$2.50$

- So the sender ended up with  $\$2.95 + \$2.50 = \$5.45$
- I ended up with  $\$6.15 - \$2.50 = \$3.65$

The sender in Round 1 showed moderate trust by sending 41% of their endowment, and I reciprocated with a return that was about 41% of what I received. This established a pattern of moderate cooperation.

Now I'm the sender. I need to decide how much to send.

Key considerations:

1. The previous sender sent 41% and got a decent return
2. I should signal trust but also be strategic
3. There's communication noise mentioned, so amounts may differ from what's actually sent
4. I want to maximize my total earnings

If I send \$2 (40%):

- Receiver gets \$6
- If they return similarly to how I did (~41%), they might return \$2.50
- I'd have:  $\$3 + \$2.50 = \$5.50$

If I send \$2.50 (50%):

- Receiver gets \$7.50
- If they return ~40%, they might return \$3
- I'd have:  $\$2.50 + \$3 = \$5.50$

If I send \$3 (60%):

- Receiver gets \$9
- If they return ~40%, they might return \$3.60
- I'd have:  $\$2 + \$3.60 = \$5.60$

If I send \$1.50 (30%):

...

**usage:**

```
{
  "input_tokens": 644,
  "output_tokens": 691,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.649999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 7 role=assistant**

**content:**

The sender sent \$0 this round, so I received \$0. Since I have nothing to return, my only option is:

```
{'return': 0}
```

**reasoning:**

The sender sent me \$0 this round, so I received \$0 after multiplication.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust me
2. They're being strategic
3. There might be communication noise mentioned in the instructions

Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

This is straightforward - there's nothing to return.

**usage:**

```
{
  "input_tokens": 851,
  "output_tokens": 148,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.6499999999999995.

How much do you send to the receiver this round? (0-5)

**Message 9 role=assistant**

**content:**

Looking at the pattern so far:

- Round 1: I cooperated well as receiver, returning \$2.50
- Round 2: I sent \$2.00 but apparently received nothing back (my earnings suggest \$0 return)
- Round 3: Sender sent me \$0

There's been a breakdown in cooperation, though the note about communication noise means perceptions might differ from reality.

I'll send a moderate amount to signal willingness to cooperate while being cautious given recent experiences. Sending \$1.50 allows me to:

- Test if cooperation can be rebuilt
- Limit potential losses if trust isn't reciprocated
- Keep open the possibility of mutual benefit

{'send': 1.50}

**reasoning:**

Let me review what's happened:

Round 1: I was receiver, got \$6.15, returned \$2.50

Round 2: I was sender, sent \$2.00

Round 3: I was receiver, sender sent \$0, I returned \$0

My total earnings are \$6.65.

Round 3 is concerning - the sender sent me \$0. This could mean:

1. They didn't trust me to return anything
2. They're being cautious or strategic
3. There might be communication noise affecting their perception
4. They're testing cooperation levels

The instructions mentioned there may be communication noise - amounts might differ from what was actually sent/returned.

In Round 2, I sent \$2.00. I don't know what they returned yet (that's not shown in my earnings update for round 3).

Wait, let me recalculate my earnings:

- Round 1: Started with \$0, received \$6.15, returned \$2.50, kept \$3.65
- Round 2: Started with \$5, sent \$2.00, kept \$3.00 initially. Need to add what was returned to me.
- Round 3: Received \$0, returned \$0, gained \$0

Total earnings = \$6.65

From Round 1: \$3.65

From Round 3: \$0

So from Round 2:  $\$6.65 - \$3.65 - \$0 = \$3.00$

This means in Round 2, I sent \$2.00 and got back \$0. The receiver didn't return anything!

So the pattern is:

- Round 1: Sender sent me \$2.05, I returned \$2.50 (good cooperation from me)
- Round 2: I sent \$2.00, receiver returned \$0 (betrayal!)
- Round 3: Sender sent me \$0 (maybe retaliating? or cautious?)

The zero in Round 3 could be retaliation for my Round 2 behavior, or it might stem from communication distortion about what I actually returned in Round 1. Since each round likely involves different partners, the person who wins...

**usage:**

```
{  
  "input_tokens": 937,
```

```

        "output_tokens": 851,
        "reasoning_tokens": 0
    }
Message 10 role=user
    content:
        Round 5

        You are the RECEIVER this round. Your total earnings across all rounds are $11.36.
        This round, The sender sent you $0, so you received $0. How much do you return to The sender? (0-0)
Message 11 role=assistant
    content:
        The sender sent $0 this round, so I received $0. With nothing received, I have nothing to return:

        {'return': 0}
    reasoning:
        The sender sent $0 this round, so I received $0. I have nothing to return.

        This is straightforward - I can only return 0 since I received nothing.
    usage:
        {
            "input_tokens": 1158,
            "output_tokens": 82,
            "reasoning_tokens": 0
        }
Message 12 role=user
    content:
        Round 6

        You are the SENDER this round. Your total earnings across all rounds are $11.36.
        How much do you send to the receiver this round? (0-5)
Message 13 role=assistant
    content:
        Looking at the pattern:

        - Round 1: Good cooperation when I was receiver
        - Round 2: I sent $2.00, got little/nothing back
        - Round 3: Received nothing as receiver
        - Round 4: I sent $1.50, got some return (earnings increased to $11.36)
        - Round 5: Received nothing as receiver again

        Cooperation has been inconsistent, though the communication noise mentioned could be affecting perceptions on both sides. The receiver in Round 4 did return something when I sent , which is a positive signal.

        I'll send a small amount to signal continued willingness to cooperate while being cautious
        :

        {'send': 1.00}
    reasoning:
        Let me analyze the pattern of play:

        Round 1: I was RECEIVER
        - Sender sent $2.05 (41%)
        - I received $6.15
        - I returned $2.50
        - My earnings: $3.65

        Round 2: I was SENDER
        - I sent $2.00
        - My total earnings after Round 2: $6.65
        - This means I gained $3.00 in Round 2 ( $6.65 - 3.65 = 3.00$ )
        - I started with $5, sent $2, left with $3
        - If I got back nothing, I'd have $3 total, but I have $6.65 total
        - Actually wait, let me recalculate. After Round 1, I had $3.65
        - In Round 2 as sender, I get $5 new endowment, send $2, keep $3
        - So I should have  $3.65 + \$3 = \$6.65$  if I got nothing back
        - My total is $6.65, so I got $0 back in Round 2

```



Round 3: I was RECEIVER  
- Sender sent \$0  
- I received \$0  
- I returned \$0  
- Total earnings stayed at \$6.65

Round 4: I was SENDER  
- I sent \$1.50  
- My total earnings after Round 4: \$6.65  
- Wait, that doesn't make sense. Let me check again.  
- Before Round 4, I had \$6.65  
- In Round 4, I get new \$5 endowment, send \$1.50, keep \$3.50  
- If I got nothing back, I'd have \$6.65 + \$3.50 = \$10.15  
- But my total after Round 4 should be reflected in what's stated before Round 5  
- The prompt shows \$11.36 total before Round 5, so after Round 4 I had \$11.36  
- That's a gain of \$4.71 from the \$6.65 I started with  
- Since I kept \$3.50 from my endowment, the receiver must have returned \$1.21

Round 5: I was RECEIVER  
- Sender sent \$0  
- I received \$0  
- I returned \$0  
- My total earnings stayed at \$11.36

Now for Round 6: I'm SENDER again  
- My total is still \$11.36

Looking at the pattern, cooperation has been sporadic. Round 1 showed genuine reciprocity when the sender trusted me with 41% and I returned generously. But Round 2 went nowhere- I sent 40% and got nothing back. Round 3 and 5 both had senders sending nothing. Round 4 was the only other time I sent (30%), and while I did get \$1.21 back, that's only 27% reciprocation on what they received- quite weak. The communication issues mentioned might be playing a role in this inconsistency....

**usage:**

```
{  
  "input_tokens": 1237,  
  "output_tokens": 1167,  
  "reasoning_tokens": 0  
}
```

**Message 14 role=user**

**content:**

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.75.  
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

The sender sent \$0 this round, so I received \$0. With nothing to return:

```
{'return': 0}
```

**Reasoning:**

The sender sent \$0 this round, so I received \$0. I have nothing to return.

**Usage:**

```
{  
  "input_tokens": 1452,  
  "output_tokens": 59,  
  "reasoning_tokens": 0  
}
```

## ROUND 8

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

**Pairings:**

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_2 (Briar) sent \$2; received \$6; Agent\_1 (Aster) returned \$1.5; payoffs Agent\_2 (Briar)=\$4.5, Agent\_1 (Aster)=\$4.5; noise/visible: sent visible=\$0.99, received visible=\$2.97, returned visible=\$1.62

Action validation: {"returned": {"amount": 1.5, "clamped": false, "max": 6.0, "min": 0, "raw": 1.5}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

```

Balances after round: {
  "Agent_1": 30.4,
  "Agent_2": 38.6
}
Visible balances after round: {
  "Agent_1": 18.71,
  "Agent_2": 20.37
}

```

## LLM CALLS

### GAME - Agent\_2 - Sender - dyad\_1

```

Timestamp: 2026-08-12T09:54:12
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {

```

```

  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}

```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.64999999999999995.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at Round 1, the sender showed moderate trust by sending 41% of their endowment, and I reciprocated with a fair return. This established a cooperative pattern.

Now as the sender, I'll maintain this level of trust and cooperation. Sending a moderate amount signals trust while managing risk.

I'll send \$2.00 (40% of my endowment):

- The receiver will get \$6.00 after the multiplier
- If they reciprocate similarly to how I did in Round 1, both of us should benefit
- This maintains the cooperative dynamic established in Round 1

{'send': 2.00}

**reasoning:**

Okay, now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was the receiver:

- Sender sent me \$2.05 (41% of their endowment)
- I received \$6.15
- I returned \$2.50
- So the sender ended up with  $\$2.95 + \$2.50 = \$5.45$
- I ended up with  $\$6.15 - \$2.50 = \$3.65$

The sender in Round 1 showed moderate trust by sending 41% of their endowment, and I reciprocated with a return that was about 41% of what I received. This established a pattern of moderate cooperation.

Now I'm the sender. I need to decide how much to send.

Key considerations:

1. The previous sender sent 41% and got a decent return
2. I should signal trust but also be strategic
3. There's communication noise mentioned, so amounts may differ from what's actually sent
4. I want to maximize my total earnings

If I send \$2 (40%):

- Receiver gets \$6
- If they return similarly to how I did (~41%), they might return \$2.50
- I'd have:  $\$3 + \$2.50 = \$5.50$

If I send \$2.50 (50%):

- Receiver gets \$7.50
- If they return ~40%, they might return \$3
- I'd have:  $\$2.50 + \$3 = \$5.50$

If I send \$3 (60%):

- Receiver gets \$9
- If they return ~40%, they might return \$3.60
- I'd have:  $\$2 + \$3.60 = \$5.60$

If I send \$1.50 (30%):

...

**usage:**

```
{
  "input_tokens": 644,
  "output_tokens": 691,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.6499999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

The sender sent \$0 this round, so I received \$0. Since I have nothing to return, my only option is:

```
{'return': 0}
```

**reasoning:**

The sender sent me \$0 this round, so I received \$0 after multiplication.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust me
2. They're being strategic
3. There might be communication noise mentioned in the instructions

Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

This is straightforward - there's nothing to return.

**usage:**

```
{
  "input_tokens": 851,
  "output_tokens": 148,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.6499999999999995.

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at the pattern so far:

- Round 1: I cooperated well as receiver, returning \$2.50
- Round 2: I sent \$2.00 but apparently received nothing back (my earnings suggest \$0 return)
- Round 3: Sender sent me \$0

There's been a breakdown in cooperation, though the note about communication noise means perceptions might differ from reality.

I'll send a moderate amount to signal willingness to cooperate while being cautious given recent experiences. Sending \$1.50 allows me to:

- Test if cooperation can be rebuilt
- Limit potential losses if trust isn't reciprocated
- Keep open the possibility of mutual benefit

```
{'send': 1.50}
```

**reasoning:**

Let me review what's happened:

Round 1: I was receiver, got \$6.15, returned \$2.50

Round 2: I was sender, sent \$2.00

Round 3: I was receiver, sender sent \$0, I returned \$0

My total earnings are \$6.65.

Round 3 is concerning - the sender sent me \$0. This could mean:

1. They didn't trust me to return anything
2. They're being cautious or strategic
3. There might be communication noise affecting their perception
4. They're testing cooperation levels

The instructions mentioned there may be communication noise - amounts might differ from what was actually sent/returned.

In Round 2, I sent \$2.00. I don't know what they returned yet (that's not shown in my earnings update for round 3).

Wait, let me recalculate my earnings:

- Round 1: Started with \$0, received \$6.15, returned \$2.50, kept \$3.65
- Round 2: Started with \$5, sent \$2.00, kept \$3.00 initially. Need to add what was returned to me.
- Round 3: Received \$0, returned \$0, gained \$0

Total earnings = \$6.65

From Round 1: \$3.65

From Round 3: \$0

So from Round 2: \$6.65 - \$3.65 - \$0 = \$3.00

This means in Round 2, I sent \$2.00 and got back \$0. The receiver didn't return anything!

So the pattern is:

- Round 1: Sender sent me \$2.05, I returned \$2.50 (good cooperation from me)
- Round 2: I sent \$2.00, receiver returned \$0 (betrayal!)
- Round 3: Sender sent me \$0 (maybe retaliating? or cautious?)

The zero in Round 3 could be retaliation for my Round 2 behavior, or it might stem from communication distortion about what I actually returned in Round 1. Since each round likely involves different partners, the person who wit...

usage:

```
{
  "input_tokens": 937,
  "output_tokens": 851,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.36.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 9 role=assistant**

**content:**

The sender sent \$0 this round, so I received \$0. With nothing received, I have nothing to return:

```
{'return': 0}
```

**reasoning:**

The sender sent \$0 this round, so I received \$0. I have nothing to return.

This is straightforward - I can only return 0 since I received nothing.

usage:

```
{
  "input_tokens": 1158,
  "output_tokens": 82,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.36.

How much do you send to the receiver this round? (0-5)

**Message 11 role=assistant**

**content:**

Looking at the pattern:

- Round 1: Good cooperation when I was receiver
- Round 2: I sent \$2.00, got little/nothing back
- Round 3: Received nothing as receiver
- Round 4: I sent \$1.50, got some return (earnings increased to \$11.36)
- Round 5: Received nothing as receiver again

Cooperation has been inconsistent, though the communication noise mentioned could be affecting perceptions on both sides. The receiver in Round 4 did return something when I sent , which is a positive signal.

I'll send a small amount to signal continued willingness to cooperate while being cautious :

```
{'send': 1.00}
```

**reasoning:**

Let me analyze the pattern of play:

Round 1: I was RECEIVER

- Sender sent \$2.05 (41%)
- I received \$6.15
- I returned \$2.50
- My earnings: \$3.65

Round 2: I was SENDER

- I sent \$2.00
- My total earnings after Round 2: \$6.65
- This means I gained \$3.00 in Round 2 ( $6.65 - 3.65 = 3.00$ )
- I started with \$5, sent \$2, left with \$3
- If I got back nothing, I'd have \$3 total, but I have \$6.65 total
- Actually wait, let me recalculate. After Round 1, I had \$3.65
- In Round 2 as sender, I get \$5 new endowment, send \$2, keep \$3
- So I should have  $\$3.65 + \$3 = \$6.65$  if I got nothing back
- My total is \$6.65, so I got \$0 back in Round 2

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- Total earnings stayed at \$6.65

Round 4: I was SENDER

- I sent \$1.50
- My total earnings after Round 4: \$6.65
- Wait, that doesn't make sense. Let me check again.
- Before Round 4, I had \$6.65
- In Round 4, I get new \$5 endowment, send \$1.50, keep \$3.50
- If I got nothing back, I'd have  $\$6.65 + \$3.50 = \$10.15$
- But my total after Round 4 should be reflected in what's stated before Round 5
- The prompt shows \$11.36 total before Round 5, so after Round 4 I had \$11.36
- That's a gain of \$4.71 from the \$6.65 I started with
- Since I kept \$3.50 from my endowment, the receiver must have returned \$1.21

Round 5: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My total earnings stayed at \$11.36

Now for Round 6: I'm SENDER again

- My total is still \$11.36

Looking at the pattern, cooperation has been sporadic. Round 1 showed genuine reciprocity when the sender trusted me with 41% and I returned generously. But Round 2 went nowhere- I sent 40% and got nothing back. Round 3 and 5 both had senders sending nothing. Round 4 was the only other time I sent (30%), and while I did get \$1.21 back, that's only 27% reciprocation on what they received- quite weak. The communication issues mentioned might be playing a role in this inconsistency...

```
usage:
{
  "input_tokens": 1237,
  "output_tokens": 1167,
  "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.75.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 13 role=assistant**

**content:**

The sender sent \$0 this round, so I received \$0. With nothing to return:

```
{'return': 0}
```

**reasoning:**

The sender sent \$0 this round, so I received \$0. I have nothing to return.

**usage:**

```
{
  "input_tokens": 1452,
  "output_tokens": 59,
  "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$15.75.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my earning progression:

- When I've been sender in recent rounds (4 & 6), I actually got good returns despite initial appearances
- Round 4: Sent \$1.50, earnings jumped significantly
- Round 6: Sent \$1.00, earnings increased again nicely
- When I'm receiver, senders send \$0

The pattern suggests that when I send, I'm getting cooperated with. I'll send a moderate amount to continue this potentially profitable pattern:

```
{'send': 2.00}
```

**Reasoning:**

Let me review the pattern:

- Round 1: I was receiver, cooperated well
- Round 2: I was sender, sent \$2.00
- Round 3: I was receiver, got \$0
- Round 4: I was sender, sent \$1.50, got some return (earnings went from \$6.65 to \$11.36, so I got about \$4.71 back from that round)
- Round 5: I was receiver, got \$0
- Round 6: I was sender, sent \$1.00, got some return (earnings went from \$11.36 to \$15.75, so I got about \$4.39 back from that round)
- Round 7: I was receiver, got \$0

So when I'm the receiver, I'm getting \$0. But when I'm the sender, I'm actually getting returns. This is interesting.

Round 4: Started with \$6.65, sent \$1.50, so I should have  $\$6.65 - \$1.50 = \$5.15$ , plus whatever return I got. I ended at \$11.36, so return was about  $\$11.36 - \$5.15 = \$6.21$ . That's actually a very generous return!

Round 6: Started with \$11.36, sent \$1.00, so I should have  $\$11.36 - \$1.00 = \$10.36$ , plus whatever return I got. I ended at \$15.75, so return was about  $\$15.75 - \$10.36 = \$5.39$ . Also a generous return!

So when I send as sender, I'm getting very good returns. But when I'm receiver, the sender sends \$0.

ds \$0.

Maybe the other player(s) are only cooperating when they're receivers, not when they're sender  
s? Or maybe there's noise affecting what they see?

Given that I've been getting good returns when I send (despite what seemed like poor cooperati  
on earlier), I should probably continue sending. ...

```
Usage: {  
  "input_tokens": 1267,  
  "output_tokens": 607,  
  "reasoning_tokens": 0  
}
```

#### **GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:54:20

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 8,  
  "task": "game",  
  "task_index": 0  
}
```

#### **Exact input messages sent to model:**

**Message 1 role=system**

##### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.

##### **GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}



MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$3.66.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

Since the sender sent me \$0, I received \$0 (even after the 3x multiplier).

With \$0 received, I have nothing to return. The only possible amount is:

```
'''json
{"return": 0}
'''
```

**reasoning:**

Let me understand the situation:

Round 1: I was SENDER and sent \$3

- I kept:  $\$5 - \$3 = \$2$
- Receiver got:  $\$3 \times 3 = \$9$
- My total earnings are \$3.66

This means I must have received \$1.66 back from the receiver in Round 1.

So:  $\$2$  (kept) +  $\$1.66$  (returned) =  $\$3.66$  ?

Round 2: I am RECEIVER

- The sender sent me \$0
- I received \$0 (since  $\$0 \times 3 = \$0$ )
- I can only return between \$0 and \$0

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

**usage:**

```
{
  "input_tokens": 567,
  "output_tokens": 245,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$3.66.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at my situation:

From Round 1, I can calculate that the receiver only returned \$1.66 out of the \$9 they received (about 18% return rate), which shows low reciprocity. However, I should note the warning about communication noise - the actual amounts may have differed.

Round 2 didn't affect my earnings since the sender sent \$0.

For Round 3, I'm facing uncertainty:

- This could be a different receiver
- Communication noise may have affected Round 1 observations

- I need to balance potential gains against the risk of poor reciprocity

I'll send a smaller amount to test this receiver's trustworthiness while limiting my downside risk:

```
'''json
{"send": 2}
'''
```

This sends \$2, which becomes \$6 for the receiver. It's enough to create mutual benefit if they reciprocate fairly, but limits my loss if they don't.

**reasoning:**

Let me analyze the situation:

Round 1: I was the SENDER and sent \$3

- My total earnings are now \$3.66
- If I sent \$3, I kept \$5 - \$3 = \$2
- The receiver got  $\$3 \times 3 = \$9$
- They returned some amount to me
- So I have:  $\$2 + \text{returned amount} = \$3.66$
- This means they returned \$1.66 to me out of the \$9 they received

That's a pretty poor return rate ( $1.66/9 \approx 18.4\%$ ). They kept \$7.34 out of \$9.

Round 2: I was the RECEIVER

- The sender sent me \$0
- So I received \$0 and returned \$0
- This didn't affect my earnings

Now in Round 3, I'm the SENDER again.

The receiver from Round 1 showed low trustworthiness by only returning \$1.66 out of \$9. However, I should note that the instructions mention "communication noise" - the amounts I see may differ from what was actually sent/returned.

Given that:

1. The previous receiver showed poor reciprocity (only returned ~18% of what they received)
2. There may be communication noise affecting what I observe
3. This might be a different receiver (it's a multi-round game, roles change)

I have a few options:

- Send \$0 to protect my earnings (risk-averse)
- Send a small amount to test this receiver
- Send a moderate amount hoping for better reciprocity

Given the poor experience in Round 1, I'll be more cautious. I'll send a smaller amount to test this receiver's trustworthiness:

I'll send \$2 this time.

**usage:**

```
{
  "input_tokens": 668,
  "output_tokens": 617,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$6.66.

This round, The sender sent you \$0.98, so you received \$2.94. How much do you return to the sender? (0-2.94)

**Message 7 role=assistant**

**content:**

Looking at this situation:

The sender sent me \$0.98, which became \$2.94 after the 3x multiplier. This created value through cooperation.

To encourage continued cooperation and show trustworthiness, I should return a fair amount

- . Let me consider the gains:
- Total value created from cooperation: \$1.96 (\$2.94 - \$0.98)
- A fair return would split these gains reasonably

If I return \$1.50:

- Sender gains: \$5 - \$0.98 + \$1.50 = \$5.52 (net gain: \$0.52)
- I gain: \$2.94 - \$1.50 = \$1.44 (net gain: \$1.44)

This gives us both positive gains from cooperating, with me gaining slightly more since I received the multiplied amount.

```
'''json
{"return": 1.50}
'''
```

#### reasoning:

Let me analyze this situation:

The sender sent me \$0.98, which was multiplied by 3x to give me \$2.94.

I need to decide how much to return. Let me think about fairness and cooperation:

1. The sender gave up \$0.98 from their \$5 endowment
2. This created \$2.94 for me
3. The total value created is \$2.94 - \$0.98 = \$1.96 in additional value due to the multiplier

A fair split might consider:

- Equal split of gains: The sender sacrificed \$0.98, I got \$2.94. If I return nothing, sender has \$4.02 and I have \$2.94. If we split the \$1.96 gain equally, I should return enough so we each gain about \$0.98.
- To give the sender a total gain of ~\$0.98, they need: \$5 - \$0.98 + return = \$5.98, so return = \$0.98
- That seems too low. Let me think differently.

Alternative approach - equal final outcomes:

- Sender started with \$5, spent \$0.98, has \$4.02 + return
- I started with \$0, got \$2.94, keep \$2.94 - return
- For equal outcomes: \$4.02 + return = \$2.94 - return
- 2\*return = \$2.94 - \$4.02 = -\$1.08
- This gives negative, which doesn't make sense

Let me think about reciprocity:

- The sender trusted me with \$0.98
- This created value of \$1.96 (the difference between 3x multiplier and original)
- A reasonable return might be to give back enough that we both benefit from the cooperation
- Testing different return amounts: returning \$1.47 would give us nearly equal final gains (\$0.49 for them, \$1.47 for me), while \$1.50 provides them \$0.52 in gains. Even \$1.20 leaves them with only \$0.22 in gains while I keep \$1.74....

#### usage:

```
{
  "input_tokens": 938,
  "output_tokens": 1003,
  "reasoning_tokens": 0
}
```

Message 8 role=user

#### content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$8.1.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

#### content:

Looking at my experience so far:

From my earnings progression:

- Round 3: I sent \$2, but my earnings only increased by \$3 (the \$3 I kept), suggesting I got \$0 back
- Round 4: I received \$2.94 and returned \$1.50, showing goodwill

The receivers I've encountered have shown poor reciprocity (0% and 18% return rates, though communication noise may be a factor). However, these might be different people each round.

Given this pattern, I need to be more cautious while still leaving room for potential cooperation. The 3x multiplier means even modest returns can be profitable.

I'll send a smaller amount to limit risk while testing if this receiver will reciprocate:

```
'''json
{"send": 1}
'''
```

This sends \$1, creating \$3 for the receiver. If they return even \$1-1.50, we both benefit. If they don't reciprocate, my loss is limited to \$1.

**reasoning:**

Let me track my earnings:

- Round 1: I sent \$3, which should have given me back something. My total after all rounds up to Round 2 was \$3.66. So I kept \$5 - \$3 = \$2, plus whatever was returned = \$3.66. So I received \$1.66 back from round 1.
- Round 2: I was receiver, sender sent \$0, so no change
- Round 3: I sent (need to calculate). Total after round 3 was still \$3.66? No wait, the prompt says "Your total earnings across all rounds are \$3.66" in Round 3. Let me recalculate.

Actually, let me track this more carefully:

- Round 1: I was sender, sent \$3. Total earnings after rounds 1-2 was \$3.66
- Round 2: I was receiver, got \$0. Total after rounds 1-2 was \$3.66
- Round 3: I was sender, sent \$2. Total after rounds 1-3 was \$6.66
- Round 4: I was receiver, got \$2.94, returned \$1.50. Total after rounds 1-4 was \$8.10

So in Round 3, I went from \$3.66 to \$6.66 = +\$3.00

I sent \$2, so I kept \$3 from my endowment

To gain \$3 total, I needed the return to be \$0

So the receiver in Round 3 returned \$0! That's disappointing.

In Round 4, the sender sent me \$0.98 (became \$2.94). I returned \$1.50. I gained \$2.94 - \$1.50 = \$1.44, which matches: \$6.66 + \$1.44 = \$8.10 ?

So my experience:

- Round 1: Sent \$3 (became \$9), got back \$1.66 (18% return)
- Round 3: Sent \$2 (became \$6), got back \$0 (0% return)
- Round 4: Received \$0.98 (became \$2.94), returned \$1.50

The reciprocity has been weak overall. Still, the 3x multiplier means even modest returns create value. I'm torn between sending nothing to protect myself and sending a small amount to test whether cooperation might work. If I send \$0, I get nothing back. A small send could still be worthwhile if receivers return even 20-30%....

**usage:**

```
{
  "input_tokens": 1195,
  "output_tokens": 992,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.91.

This round, The sender sent you \$0.36, so you received \$1.08. How much do you return to the sender? (0-1.08)

**Message 11 role=assistant**

**content:**

Looking at this situation:

The sender sent me \$0.36, which became \$1.08 after the 3x multiplier. This is a very small amount, suggesting they have low trust - likely from poor past experiences.

To demonstrate trustworthiness and potentially build cooperation:

Total value created from cooperation:  $\$1.08 - \$0.36 = \$0.72$

If I return \$0.60:

- Sender gains:  $\$5 - \$0.36 + \$0.60 = \$5.24$  (net +\$0.24)
- I gain:  $\$1.08 - \$0.60 = \$0.48$  (net +\$0.48)

This shows strong reciprocity (~56% return rate) which could encourage more trust and larger exchanges in future rounds, benefiting both of us.

```
'''json
{"return": 0.60}
'''
```

#### reasoning:

Let me analyze this situation:

The sender sent me \$0.36, which became \$1.08 after the 3x multiplier.

This is a very small amount - they're being quite cautious, likely due to past experiences with low reciprocity.

Let me think about what's fair:

- Total value created:  $\$1.08 - \$0.36 = \$0.72$
- A fair split of gains might be around half each

If I return \$0.54:

- Sender:  $\$5 - \$0.36 + \$0.54 = \$5.18$  (gain of \$0.18)
- Me:  $\$1.08 - \$0.54 = \$0.54$  (gain of \$0.54)

This gives us both positive gains, with me gaining more since I received the multiplied amount.

Alternatively, if I return \$0.70:

- Sender:  $\$5 - \$0.36 + \$0.70 = \$5.34$  (gain of \$0.34)
- Me:  $\$1.08 - \$0.70 = \$0.38$  (gain of \$0.38)

This is closer to an equal split of the total value created.

Given that:

1. The sender sent a very small amount (showing low trust)
2. I want to demonstrate trustworthiness to encourage future cooperation
3. Even a modest return shows good faith
4. The amounts are small, so being generous could pay off

I'll return about 50-65% of what I received, which shows strong reciprocity and might encourage the sender to trust more in future rounds.

Let me return \$0.60:

...

#### usage:

```
{
  "input_tokens": 1485,
  "output_tokens": 669,
  "reasoning_tokens": 0
}
```

Message 12 role=user

#### content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.39.  
How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

#### content:

Looking at my track record:

As SENDER:

- Round 1: Sent \$3, got back \$1.66 (low reciprocity)
- Round 3: Sent \$2, got back \$0 (no reciprocity)
- Round 5: Sent \$1, got back \$0.81 (best reciprocity rate ~27%)

As RECEIVER, I've been returning 50-56%, demonstrating trustworthiness.

Interestingly, my smallest send (Round 5) got the best return rate. This could be due to:

- Communication noise making small amounts appear larger
- Building reputation from my fair behavior as receiver
- Different receivers with varying trustworthiness

I'll send a moderate amount to test if cooperation is improving:

```
'''json
{"send": 2}
'''
```

This sends \$2 (becomes \$6). If reciprocity continues improving, we both benefit. If not, my loss is limited.

**reasoning:**

Let me track my earnings:

- Round 1 (SENDER): I sent \$3, ended with \$3.66. So I kept \$2 and got back \$1.66
- Round 2 (RECEIVER): Sender sent \$0, I got \$0. No change to earnings.
- Round 3 (SENDER): Started at \$3.66. I sent \$2. Ended at \$6.66, so gained \$3. This means I kept \$3 and got back \$0.
- Round 4 (RECEIVER): Started at \$6.66. Received \$2.94, returned \$1.50. Gained \$1.44 (\$2.94-\$1.50). Ended at \$8.1.
- Round 5 (SENDER): Started at \$8.1. I sent \$1. Ended at \$12.91, so gained \$4.81. I kept \$4, so I got back \$0.81.
- Round 6 (RECEIVER): Started at \$12.91. Received \$1.08, returned \$0.60. Gained \$0.48 (\$1.08-\$0.60). Ended at \$13.39.
- Round 7 (SENDER): Starting at \$13.39

Looking at patterns when I was SENDER:

- Round 1: Sent \$3, got back \$1.66 (18.4% of the \$9 received)
- Round 3: Sent \$2, got back \$0 (0% of the \$6 received)
- Round 5: Sent \$1, got back \$0.81 (27% of the \$3 received)

Interesting - Round 5 showed better reciprocity (27% return) when I sent less. And I've been showing good reciprocity when I'm a receiver (50% in Round 4, 56% in Round 6).

The pattern suggests:

1. Different receivers each round

...

**usage:**

```
{
  "input_tokens": 1735,
  "output_tokens": 797,
  "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$17.240000000000002.

This round, The sender sent you \$0.99, so you received \$2.9699999999999998. How much do you return to The sender? (0-2.9699999999999998)

**Assistant output:**

Looking at this situation:

The sender sent me \$0.99, which became approximately \$2.97 after the 3x multiplier.

I've been maintaining a consistent pattern of reciprocity as a receiver (50-56% return rates in previous rounds), which helps build trust and encourages future cooperation.

For this round:

- Value created from cooperation: \$2.97 - \$0.99 = \$1.98

If I return \$1.50:

- Sender gains: \$5 - \$0.99 + \$1.50 = \$5.51 (net +\$0.51)
- I gain: \$2.97 - \$1.50 = \$1.47 (net +\$1.47)

This maintains my fair reciprocity pattern (~50% return rate) and ensures we both benefit from

cooperation.

```
'''json
{"return": 1.50}
'''
```

#### Reasoning:

The sender sent me \$0.99, which became approximately \$2.97 after the 3x multiplier.

Looking at what I've been doing as a receiver:

- Round 2: Received \$0, returned \$0 (nothing to return)
- Round 4: Received \$2.94, returned \$1.50 (~51% return rate)
- Round 6: Received \$1.08, returned \$0.60 (~56% return rate)

I've been consistently fair as a receiver, returning about 50-56% of what I receive. This helps build trust and encourages cooperation.

For this round:

- Sender sent: \$0.99
- I received: ~\$2.97
- Value created:  $\$2.97 - \$0.99 = \$1.98$

A fair return would be around 50-55% of what I received, which would be \$1.48-\$1.63.

If I return \$1.50:

- Sender gets:  $\$5 - \$0.99 + \$1.50 = \$5.51$  (net gain: \$0.51)
- I get:  $\$2.97 - \$1.50 = \$1.47$  (net gain: \$1.47)

This maintains my pattern of reciprocity and shares the gains fairly, with me getting slightly more since I received the multiplied amount.

```
Usage: {
  "input_tokens": 1859,
  "output_tokens": 535,
  "reasoning_tokens": 0
}
```

#### ROUND 9

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

#### Pairings:

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_1 (Aster) sent \$1; received \$3; Agent\_2 (Briar) returned \$0.3; payoffs Agent\_1 (Aster)=\$4.3, Agent\_2 (Briar)=\$2.7; noise/visible: sent visible=\$0.21, received visible=\$0.63, returned visible=\$0  
Action validation: {"returned": {"amount": 0.3, "clamped": false, "max": 3.0, "min": 0, "raw": 0.3}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

```
"Agent_1": 34.699999999999996,
"Agent_2": 41.300000000000004
```

}

Visible balances after round: {

```
"Agent_1": 22.71,
"Agent_2": 20.7
```

}

#### LLM CALLS

#### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-12T09:54:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
```

```

    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$3.66.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at my situation:

From Round 1, I can calculate that the receiver only returned \$1.66 out of the \$9 they received (about 18% return rate), which shows low reciprocity. However, I should note the warning about communication noise - the actual amounts may have differed.



Round 2 didn't affect my earnings since the sender sent \$0.

For Round 3, I'm facing uncertainty:

- This could be a different receiver
- Communication noise may have affected Round 1 observations
- I need to balance potential gains against the risk of poor reciprocity

I'll send a smaller amount to test this receiver's trustworthiness while limiting my downside risk:

```
'''json
{"send": 2}
'''
```

This sends \$2, which becomes \$6 for the receiver. It's enough to create mutual benefit if they reciprocate fairly, but limits my loss if they don't.

**reasoning:**

Let me analyze the situation:

Round 1: I was the SENDER and sent \$3

- My total earnings are now \$3.66
- If I sent \$3, I kept \$5 - \$3 = \$2
- The receiver got  $\$3 \times 3 = \$9$
- They returned some amount to me
- So I have:  $\$2 + \text{returned amount} = \$3.66$
- This means they returned \$1.66 to me out of the \$9 they received

That's a pretty poor return rate ( $1.66/9 \approx 18.4\%$ ). They kept \$7.34 out of \$9.

Round 2: I was the RECEIVER

- The sender sent me \$0
- So I received \$0 and returned \$0
- This didn't affect my earnings

Now in Round 3, I'm the SENDER again.

The receiver from Round 1 showed low trustworthiness by only returning \$1.66 out of \$9. However, I should note that the instructions mention "communication noise" - the amounts I see may differ from what was actually sent/returned.

Given that:

1. The previous receiver showed poor reciprocity (only returned ~18% of what they received)
2. There may be communication noise affecting what I observe
3. This might be a different receiver (it's a multi-round game, roles change)

I have a few options:

- Send \$0 to protect my earnings (risk-averse)
- Send a small amount to test this receiver
- Send a moderate amount hoping for better reciprocity

Given the poor experience in Round 1, I'll be more cautious. I'll send a smaller amount to test this receiver's trustworthiness:

I'll send \$2 this time.

**usage:**

```
{
  "input_tokens": 668,
  "output_tokens": 617,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$6.66.

This round, The sender sent you \$0.98, so you received \$2.94. How much do you return to the sender? (0-2.94)

**Message 5 role=assistant**

**content:**

Looking at this situation:

The sender sent me \$0.98, which became \$2.94 after the 3x multiplier. This created value through cooperation.

To encourage continued cooperation and show trustworthiness, I should return a fair amount. Let me consider the gains:

- Total value created from cooperation: \$1.96 (\$2.94 - \$0.98)
- A fair return would split these gains reasonably

If I return \$1.50:

- Sender gains: \$5 - \$0.98 + \$1.50 = \$5.52 (net gain: \$0.52)
- I gain: \$2.94 - \$1.50 = \$1.44 (net gain: \$1.44)

This gives us both positive gains from cooperating, with me gaining slightly more since I received the multiplied amount.

```
'''json
{"return": 1.50}
'''
```

**reasoning:**

Let me analyze this situation:

The sender sent me \$0.98, which was multiplied by 3x to give me \$2.94.

I need to decide how much to return. Let me think about fairness and cooperation:

1. The sender gave up \$0.98 from their \$5 endowment
2. This created \$2.94 for me
3. The total value created is \$2.94 - \$0.98 = \$1.96 in additional value due to the multiplier

A fair split might consider:

- Equal split of gains: The sender sacrificed \$0.98, I got \$2.94. If I return nothing, sender has \$4.02 and I have \$2.94. If we split the \$1.96 gain equally, I should return enough so we each gain about \$0.98.
- To give the sender a total gain of ~\$0.98, they need: \$5 - \$0.98 + return = \$5.98, so return = \$0.98
- That seems too low. Let me think differently.

Alternative approach - equal final outcomes:

- Sender started with \$5, spent \$0.98, has \$4.02 + return
- I started with \$0, got \$2.94, keep \$2.94 - return
- For equal outcomes: \$4.02 + return = \$2.94 - return
- $2 \times \text{return} = \$2.94 - \$4.02 = -\$1.08$
- This gives negative, which doesn't make sense

Let me think about reciprocity:

- The sender trusted me with \$0.98
- This created value of \$1.96 (the difference between 3x multiplier and original)
- A reasonable return might be to give back enough that we both benefit from the cooperation
- Testing different return amounts: returning \$1.47 would give us nearly equal final gains (\$0.49 for them, \$1.47 for me), while \$1.50 provides them \$0.52 in gains. Even \$1.20 leaves them with only \$0.22 in gains while I keep \$1.74....

**usage:**

```
{
  "input_tokens": 938,
  "output_tokens": 1003,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$8.1.

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at my experience so far:

From my earnings progression:

- Round 3: I sent \$2, but my earnings only increased by \$3 (the \$3 I kept), suggesting I got \$0 back
- Round 4: I received \$2.94 and returned \$1.50, showing goodwill

The receivers I've encountered have shown poor reciprocity (0% and 18% return rates, though communication noise may be a factor). However, these might be different people each round.

Given this pattern, I need to be more cautious while still leaving room for potential cooperation. The 3x multiplier means even modest returns can be profitable.

I'll send a smaller amount to limit risk while testing if this receiver will reciprocate:

```
'''json
{"send": 1}
'''
```

This sends \$1, creating \$3 for the receiver. If they return even \$1-1.50, we both benefit. If they don't reciprocate, my loss is limited to \$1.

**reasoning:**

Let me track my earnings:

- Round 1: I sent \$3, which should have given me back something. My total after all rounds up to Round 2 was \$3.66. So I kept \$5 - \$3 = \$2, plus whatever was returned = \$3.66. So I received \$1.66 back from round 1.
- Round 2: I was receiver, sender sent \$0, so no change
- Round 3: I sent (need to calculate). Total after round 3 was still \$3.66? No wait, the prompt says "Your total earnings across all rounds are \$3.66" in Round 3. Let me recalculate.

Actually, let me track this more carefully:

- Round 1: I was sender, sent \$3. Total earnings after rounds 1-2 was \$3.66
- Round 2: I was receiver, got \$0. Total after rounds 1-2 was \$3.66
- Round 3: I was sender, sent \$2. Total after rounds 1-3 was \$6.66
- Round 4: I was receiver, got \$2.94, returned \$1.50. Total after rounds 1-4 was \$8.10

So in Round 3, I went from \$3.66 to \$6.66 = +\$3.00

I sent \$2, so I kept \$3 from my endowment

To gain \$3 total, I needed the return to be \$0

So the receiver in Round 3 returned \$0! That's disappointing.

In Round 4, the sender sent me \$0.98 (became \$2.94). I returned \$1.50. I gained \$2.94 - \$1.50 = \$1.44, which matches: \$6.66 + \$1.44 = \$8.10 ?

So my experience:

- Round 1: Sent \$3 (became \$9), got back \$1.66 (18% return)
- Round 3: Sent \$2 (became \$6), got back \$0 (0% return)
- Round 4: Received \$0.98 (became \$2.94), returned \$1.50

The reciprocity has been weak overall. Still, the 3x multiplier means even modest returns create value. I'm torn between sending nothing to protect myself and sending a small amount to test whether cooperation might work. If I send \$0, I get nothing back. A small send could still be worthwhile if receivers return even 20-30%...

**usage:**

```
{
  "input_tokens": 1195,
  "output_tokens": 992,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.91.

This round, The sender sent you \$0.36, so you received \$1.08. How much do you return to the sender? (0-1.08)

**Message 9 role=assistant**

**content:**

Looking at this situation:

The sender sent me \$0.36, which became \$1.08 after the 3x multiplier. This is a very small amount, suggesting they have low trust - likely from poor past experiences.

To demonstrate trustworthiness and potentially build cooperation:

Total value created from cooperation:  $\$1.08 - \$0.36 = \$0.72$

If I return \$0.60:

- Sender gains:  $\$5 - \$0.36 + \$0.60 = \$5.24$  (net +\$0.24)
- I gain:  $\$1.08 - \$0.60 = \$0.48$  (net +\$0.48)

This shows strong reciprocity (~56% return rate) which could encourage more trust and larger exchanges in future rounds, benefiting both of us.

```
'''json
{"return": 0.60}
'''
```

**reasoning:**

Let me analyze this situation:

The sender sent me \$0.36, which became \$1.08 after the 3x multiplier.

This is a very small amount - they're being quite cautious, likely due to past experiences with low reciprocity.

Let me think about what's fair:

- Total value created:  $\$1.08 - \$0.36 = \$0.72$
- A fair split of gains might be around half each

If I return \$0.54:

- Sender:  $\$5 - \$0.36 + \$0.54 = \$5.18$  (gain of \$0.18)
- Me:  $\$1.08 - \$0.54 = \$0.54$  (gain of \$0.54)

This gives us both positive gains, with me gaining more since I received the multiplied amount.

Alternatively, if I return \$0.70:

- Sender:  $\$5 - \$0.36 + \$0.70 = \$5.34$  (gain of \$0.34)
- Me:  $\$1.08 - \$0.70 = \$0.38$  (gain of \$0.38)

This is closer to an equal split of the total value created.

Given that:

1. The sender sent a very small amount (showing low trust)
2. I want to demonstrate trustworthiness to encourage future cooperation
3. Even a modest return shows good faith
4. The amounts are small, so being generous could pay off

I'll return about 50-65% of what I received, which shows strong reciprocity and might encourage the sender to trust more in future rounds.

Let me return \$0.60:

...

**usage:**

```
{
  "input_tokens": 1485,
  "output_tokens": 669,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.39.

How much do you send to the receiver this round? (0-5)

**Message 11 role=assistant**

**content:**

Looking at my track record:

As SENDER:

- Round 1: Sent \$3, got back \$1.66 (low reciprocity)
- Round 3: Sent \$2, got back \$0 (no reciprocity)
- Round 5: Sent \$1, got back \$0.81 (best reciprocity rate ~27%)

As RECEIVER, I've been returning 50-56%, demonstrating trustworthiness.

Interestingly, my smallest send (Round 5) got the best return rate. This could be due to:

- Communication noise making small amounts appear larger
- Building reputation from my fair behavior as receiver
- Different receivers with varying trustworthiness

I'll send a moderate amount to test if cooperation is improving:

```
'''json
{"send": 2}
'''
```

This sends \$2 (becomes \$6). If reciprocity continues improving, we both benefit. If not, my loss is limited.

**reasoning:**

Let me track my earnings:

- Round 1 (SENDER): I sent \$3, ended with \$3.66. So I kept \$2 and got back \$1.66
- Round 2 (RECEIVER): Sender sent \$0, I got \$0. No change to earnings.
- Round 3 (SENDER): Started at \$3.66. I sent \$2. Ended at \$6.66, so gained \$3. This means I kept \$3 and got back \$0.
- Round 4 (RECEIVER): Started at \$6.66. Received \$2.94, returned \$1.50. Gained \$1.44 (\$2.94-\$1.50). Ended at \$8.1.
- Round 5 (SENDER): Started at \$8.1. I sent \$1. Ended at \$12.91, so gained \$4.81. I kept \$4, so I got back \$0.81.
- Round 6 (RECEIVER): Started at \$12.91. Received \$1.08, returned \$0.60. Gained \$0.48 (\$1.08-\$0.60). Ended at \$13.39.
- Round 7 (SENDER): Starting at \$13.39

Looking at patterns when I was SENDER:

- Round 1: Sent \$3, got back \$1.66 (18.4% of the \$9 received)
- Round 3: Sent \$2, got back \$0 (0% of the \$6 received)
- Round 5: Sent \$1, got back \$0.81 (27% of the \$3 received)

Interesting - Round 5 showed better reciprocity (27% return) when I sent less. And I've been showing good reciprocity when I'm a receiver (50% in Round 4, 56% in Round 6).

The pattern suggests:

1. Different receivers each round

...

**usage:**

```
{
  "input_tokens": 1735,
  "output_tokens": 797,
  "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$17.240000000000002.

This round, The sender sent you \$0.99, so you received \$2.9699999999999998. How much do you return to The sender? (0-2.9699999999999998)

**Message 13 role=assistant**

**content:**

Looking at this situation:

The sender sent me \$0.99, which became approximately \$2.97 after the 3x multiplier.

I've been maintaining a consistent pattern of reciprocity as a receiver (50-56% return rates in previous rounds), which helps build trust and encourages future cooperation.

For this round:

- Value created from cooperation: \$2.97 - \$0.99 = \$1.98

If I return \$1.50:  
- Sender gains:  $\$5 - \$0.99 + \$1.50 = \$5.51$  (net +\$0.51)  
- I gain:  $\$2.97 - \$1.50 = \$1.47$  (net +\$1.47)

This maintains my fair reciprocity pattern (~50% return rate) and ensures we both benefit from cooperation.

```
'''json
{"return": 1.50}
'''
```

**reasoning:**

The sender sent me \$0.99, which became approximately \$2.97 after the 3x multiplier.

Looking at what I've been doing as a receiver:

- Round 2: Received \$0, returned \$0 (nothing to return)
- Round 4: Received \$2.94, returned \$1.50 (~51% return rate)
- Round 6: Received \$1.08, returned \$0.60 (~56% return rate)

I've been consistently fair as a receiver, returning about 50-56% of what I receive. This helps build trust and encourages cooperation.

For this round:

- Sender sent: \$0.99
- I received: ~\$2.97
- Value created:  $\$2.97 - \$0.99 = \$1.98$

A fair return would be around 50-55% of what I received, which would be \$1.48-\$1.63.

If I return \$1.50:  
- Sender gets:  $\$5 - \$0.99 + \$1.50 = \$5.51$  (net gain: \$0.51)  
- I get:  $\$2.97 - \$1.50 = \$1.47$  (net gain: \$1.47)

This maintains my pattern of reciprocity and shares the gains fairly, with me getting slightly more since I received the multiplied amount.

**usage:**

```
{
  "input_tokens": 1859,
  "output_tokens": 535,
  "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$18.71.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my track record:

As SENDER:

- Round 1: Sent \$3 ? got \$1.66 back (18% return)
- Round 3: Sent \$2 ? got \$0 back (0% return)
- Round 5: Sent \$1 ? got \$0.81 back (27% return - best!)
- Round 7: Sent \$2 ? got \$0.85 back (14% return)

Pattern: Despite my fair behavior as a receiver (50-56% returns), receivers have consistently shown poor reciprocity. Interestingly, my smallest send (\$1) yielded the best return rate.

Given this pattern and considering communication noise may be affecting perceptions, I'll send a smaller amount:

```
'''json
{"send": 1}
'''
```

This limits my downside risk to \$1 while still creating \$3 for the receiver. If they reciprocate even modestly, we both gain. Based on past rounds, smaller sends seem to perform better.

**Reasoning:**

Let me track my earnings:

- Started Round 3 with \$3.66
- Round 3 (SENDER): Sent \$2, kept \$3, got back ~\$0 ? total \$6.66
- Round 4 (RECEIVER): Got \$2.94, returned \$1.50, kept \$1.44 ? total \$8.10
- Round 5 (SENDER): Sent \$1, kept \$4, got back \$0.81 ? total \$12.91
- Round 6 (RECEIVER): Got \$1.08, returned \$0.60, kept \$0.48 ? total \$13.39
- Round 7 (SENDER): Sent \$2, kept \$3, got back ???
  - Started with \$13.39, now at \$17.24, so gained \$3.85
  - That means I got back \$0.85 (kept \$3 + returned \$0.85 = \$3.85)
- Round 8 (RECEIVER): Got \$2.97, returned \$1.50, kept \$1.47 ? total \$18.71

So in Round 7, I sent \$2 and got back only \$0.85 (return rate ~14%).

My experience as SENDER:

- Round 1: Sent \$3 ? got \$1.66 back (return rate 18%)
- Round 3: Sent \$2 ? got \$0 back (return rate 0%)
- Round 5: Sent \$1 ? got \$0.81 back (return rate 27%)
- Round 7: Sent \$2 ? got \$0.85 back (return rate 14%)

The pattern shows:

- Receivers are generally not reciprocating well
- My smaller sends (\$1) actually got better percentage returns
- Even though I've been a fair receiver (returning 50-56%), this hasn't improved sender experiences

Given this consistent poor reciprocity from receivers, and considering communication noise may be affecting things, I should:

...

```
Usage: {
  "input_tokens": 1997,
  "output_tokens": 746,
  "reasoning_tokens": 0
}
```

#### GAME - Agent\_2 - Receiver - dyad\_1

Timestamp: 2026-08-12T09:54:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.6499999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

The sender sent \$0 this round, so I received \$0. Since I have nothing to return, my only option is:

```
{'return': 0}
```

**reasoning:**

The sender sent me \$0 this round, so I received \$0 after multiplication.

This is interesting - the sender chose not to send anything. This could mean:

1. They don't trust me
2. They're being strategic
3. There might be communication noise mentioned in the instructions

Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

This is straightforward - there's nothing to return.

**usage:**

```
{
  "input_tokens": 851,
  "output_tokens": 148,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.6499999999999995.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**



**content:**

Looking at the pattern so far:

- Round 1: I cooperated well as receiver, returning \$2.50
- Round 2: I sent \$2.00 but apparently received nothing back (my earnings suggest \$0 return)
- Round 3: Sender sent me \$0

There's been a breakdown in cooperation, though the note about communication noise means perceptions might differ from reality.

I'll send a moderate amount to signal willingness to cooperate while being cautious given recent experiences. Sending \$1.50 allows me to:

- Test if cooperation can be rebuilt
- Limit potential losses if trust isn't reciprocated
- Keep open the possibility of mutual benefit

```
{'send': 1.50}
```

**reasoning:**

Let me review what's happened:

Round 1: I was receiver, got \$6.15, returned \$2.50

Round 2: I was sender, sent \$2.00

Round 3: I was receiver, sender sent \$0, I returned \$0

My total earnings are \$6.65.

Round 3 is concerning - the sender sent me \$0. This could mean:

1. They didn't trust me to return anything
2. They're being cautious or strategic
3. There might be communication noise affecting their perception
4. They're testing cooperation levels

The instructions mentioned there may be communication noise - amounts might differ from what was actually sent/returned.

In Round 2, I sent \$2.00. I don't know what they returned yet (that's not shown in my earnings update for round 3).

Wait, let me recalculate my earnings:

- Round 1: Started with \$0, received \$6.15, returned \$2.50, kept \$3.65
- Round 2: Started with \$5, sent \$2.00, kept \$3.00 initially. Need to add what was returned to me.
- Round 3: Received \$0, returned \$0, gained \$0

Total earnings = \$6.65

From Round 1: \$3.65

From Round 3: \$0

So from Round 2:  $\$6.65 - \$3.65 - \$0 = \$3.00$

This means in Round 2, I sent \$2.00 and got back \$0. The receiver didn't return anything!

So the pattern is:

- Round 1: Sender sent me \$2.05, I returned \$2.50 (good cooperation from me)
- Round 2: I sent \$2.00, receiver returned \$0 (betrayal!)
- Round 3: Sender sent me \$0 (maybe retaliating? or cautious?)

The zero in Round 3 could be retaliation for my Round 2 behavior, or it might stem from communication distortion about what I actually returned in Round 1. Since each round likely involves different partners, the person who withdrew...

**usage:**

```
{
  "input_tokens": 937,
  "output_tokens": 851,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.36.  
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 7 role=assistant**

**content:**

The sender sent \$0 this round, so I received \$0. With nothing received, I have nothing to return:

```
{'return': 0}
```

**reasoning:**

The sender sent \$0 this round, so I received \$0. I have nothing to return.

This is straightforward - I can only return 0 since I received nothing.

**usage:**

```
{
  "input_tokens": 1158,
  "output_tokens": 82,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.36.  
How much do you send to the receiver this round? (0-5)

**Message 9 role=assistant**

**content:**

Looking at the pattern:

- Round 1: Good cooperation when I was receiver
- Round 2: I sent \$2.00, got little/nothing back
- Round 3: Received nothing as receiver
- Round 4: I sent \$1.50, got some return (earnings increased to \$11.36)
- Round 5: Received nothing as receiver again

Cooperation has been inconsistent, though the communication noise mentioned could be affecting perceptions on both sides. The receiver in Round 4 did return something when I sent , which is a positive signal.

I'll send a small amount to signal continued willingness to cooperate while being cautious :

```
{'send': 1.00}
```

**reasoning:**

Let me analyze the pattern of play:

Round 1: I was RECEIVER

- Sender sent \$2.05 (41%)
- I received \$6.15
- I returned \$2.50
- My earnings: \$3.65

Round 2: I was SENDER

- I sent \$2.00
- My total earnings after Round 2: \$6.65
- This means I gained \$3.00 in Round 2 ( $6.65 - 3.65 = 3.00$ )
- I started with \$5, sent \$2, left with \$3
- If I got back nothing, I'd have \$3 total, but I have \$6.65 total
- Actually wait, let me recalculate. After Round 1, I had \$3.65
- In Round 2 as sender, I get \$5 new endowment, send \$2, keep \$3
- So I should have  $3.65 + \$3 = \$6.65$  if I got nothing back
- My total is \$6.65, so I got \$0 back in Round 2

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- Total earnings stayed at \$6.65

Round 4: I was SENDER

- I sent \$1.50
- My total earnings after Round 4: \$6.65
- Wait, that doesn't make sense. Let me check again.
- Before Round 4, I had \$6.65
- In Round 4, I get new \$5 endowment, send \$1.50, keep \$3.50
- If I got nothing back, I'd have  $\$6.65 + \$3.50 = \$10.15$
- But my total after Round 4 should be reflected in what's stated before Round 5
- The prompt shows \$11.36 total before Round 5, so after Round 4 I had \$11.36
- That's a gain of \$4.71 from the \$6.65 I started with
- Since I kept \$3.50 from my endowment, the receiver must have returned \$1.21

Round 5: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My total earnings stayed at \$11.36

Now for Round 6: I'm SENDER again

- My total is still \$11.36

Looking at the pattern, cooperation has been sporadic. Round 1 showed genuine reciprocity when the sender trusted me with 41% and I returned generously. But Round 2 went nowhere- I sent 40% and got nothing back. Round 3 and 5 both had senders sending nothing. Round 4 was the only other time I sent (30%), and while I did get \$1.21 back, that's only 27% reciprocation on what they received- quite weak. The communication issues mentioned might be playing a role in this inconsistency....

usage:

```
{
  "input_tokens": 1237,
  "output_tokens": 1167,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.75.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 11 role=assistant**

**content:**

The sender sent \$0 this round, so I received \$0. With nothing to return:

```
{'return': 0}
```

**reasoning:**

The sender sent \$0 this round, so I received \$0. I have nothing to return.

usage:

```
{
  "input_tokens": 1452,
  "output_tokens": 59,
  "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$15.75.

How much do you send to the receiver this round? (0-5)

**Message 13 role=assistant**

**content:**

Looking at my earning progression:

- When I've been sender in recent rounds (4 & 6), I actually got good returns despite initial appearances
- Round 4: Sent \$1.50, earnings jumped significantly
- Round 6: Sent \$1.00, earnings increased again nicely
- When I'm receiver, senders send \$0

The pattern suggests that when I send, I'm getting cooperated with. I'll send a moderate amount to continue this potentially profitable pattern:

```
{'send': 2.00}
```

**reasoning:**

Let me review the pattern:

- Round 1: I was receiver, cooperated well
- Round 2: I was sender, sent \$2.00
- Round 3: I was receiver, got \$0
- Round 4: I was sender, sent \$1.50, got some return (earnings went from \$6.65 to \$11.36, so I got about \$4.71 back from that round)
- Round 5: I was receiver, got \$0
- Round 6: I was sender, sent \$1.00, got some return (earnings went from \$11.36 to \$15.75, so I got about \$4.39 back from that round)
- Round 7: I was receiver, got \$0

So when I'm the receiver, I'm getting \$0. But when I'm the sender, I'm actually getting returns. This is interesting.

Round 4: Started with \$6.65, sent \$1.50, so I should have  $\$6.65 - \$1.50 = \$5.15$ , plus whatever return I got. I ended at \$11.36, so return was about  $\$11.36 - \$5.15 = \$6.21$ . That's actually a very generous return!

Round 6: Started with \$11.36, sent \$1.00, so I should have  $\$11.36 - \$1.00 = \$10.36$ , plus whatever return I got. I ended at \$15.75, so return was about  $\$15.75 - \$10.36 = \$5.39$ . Also a generous return!

So when I send as sender, I'm getting very good returns. But when I'm receiver, the sender sends \$0.

Maybe the other player(s) are only cooperating when they're receivers, not when they're senders? Or maybe there's noise affecting what they see?

Given that I've been getting good returns when I send (despite what seemed like poor cooperation earlier), I should probably continue sending. ...

**usage:**

```
{
  "input_tokens": 1267,
  "output_tokens": 607,
  "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$20.37.

This round, The sender sent you \$0.21, so you received \$0.63. How much do you return to The sender? (0-0.63)

**Assistant output:**

The sender sent me \$0.21, which tripled to \$0.63. This is the first time a sender has sent me something in several rounds!

I'll return a fair portion to encourage continued cooperation:

```
{'return': 0.30}
```

**Reasoning:**

The sender sent me \$0.21, which was multiplied by 3, so I received \$0.63.

This is the first round where a sender has sent me something other than \$0! The amount is small (\$0.21), but it's a signal of cooperation.

They sent \$0.21 out of \$5, keeping \$4.79.  
I received \$0.63.

I should return something to encourage cooperation. Since they sent a small amount, I should return a reasonable portion to signal I'm willing to cooperate.

If I return \$0.30, that's about half of what I received. That seems fair and reciprocal.

- Sender would end with:  $\$4.79 + \$0.30 = \$5.09$
- I would end with:  $\$0.63 - \$0.30 = \$0.33$

This is a positive signal that could build trust for future rounds.

Let me return \$0.30 to show I'm willing to reciprocate cooperation.

```
Usage: {
  "input_tokens": 1257,
  "output_tokens": 291,
  "reasoning_tokens": 0
}
```

## ROUND 10

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

### Pairings:

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

### Parsed game decisions:

dyad\_1: Agent\_2 (Briar) sent \$2.5; received \$7.5; Agent\_1 (Aster) returned \$0; payoffs Agent\_2 (Briar)=\$2.5, Agent\_1 (Aster)=\$7.5; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.12

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 7.5, "min": 0, "raw": 0.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent\_1": 42.199999999999996,  
"Agent\_2": 43.800000000000004

}  
Visible balances after round: {

"Agent\_1": 22.71,  
"Agent\_2": 23.32

}

## LLM CALLS

### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-12T09:54:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_1",  
"pairing": {  
 "agent\_names": {  
 "Agent\_1": "Aster",  
 "Agent\_2": "Briar"

},  
"agents": [  
 "Agent\_2",  
 "Agent\_1"

],  
"investor": "Agent\_2",  
"roles": {  
 "Agent\_1": "trustee",  
 "Agent\_2": "investor"

},  
"trustee": "Agent\_1"

},  
"role": "investor",  
"role\_label": "Sender",  
"round": 10,  
"task": "game",  
"task\_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.6499999999999995.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at the pattern so far:

- Round 1: I cooperated well as receiver, returning \$2.50
- Round 2: I sent \$2.00 but apparently received nothing back (my earnings suggest \$0 return)
- Round 3: Sender sent me \$0

There's been a breakdown in cooperation, though the note about communication noise means perceptions might differ from reality.

I'll send a moderate amount to signal willingness to cooperate while being cautious given recent experiences. Sending \$1.50 allows me to:

- Test if cooperation can be rebuilt
- Limit potential losses if trust isn't reciprocated
- Keep open the possibility of mutual benefit

{'send': 1.50}

**reasoning:**

Let me review what's happened:

Round 1: I was receiver, got \$6.15, returned \$2.50

Round 2: I was sender, sent \$2.00

Round 3: I was receiver, sender sent \$0, I returned \$0

My total earnings are \$6.65.

Round 3 is concerning - the sender sent me \$0. This could mean:

1. They didn't trust me to return anything
2. They're being cautious or strategic
3. There might be communication noise affecting their perception
4. They're testing cooperation levels

The instructions mentioned there may be communication noise - amounts might differ from what was actually sent/returned.

In Round 2, I sent \$2.00. I don't know what they returned yet (that's not shown in my earnings update for round 3).

Wait, let me recalculate my earnings:

- Round 1: Started with \$0, received \$6.15, returned \$2.50, kept \$3.65
- Round 2: Started with \$5, sent \$2.00, kept \$3.00 initially. Need to add what was returned to me.
- Round 3: Received \$0, returned \$0, gained \$0

Total earnings = \$6.65

From Round 1: \$3.65

From Round 3: \$0

So from Round 2:  $\$6.65 - \$3.65 - \$0 = \$3.00$

This means in Round 2, I sent \$2.00 and got back \$0. The receiver didn't return anything!

So the pattern is:

- Round 1: Sender sent me \$2.05, I returned \$2.50 (good cooperation from me)
- Round 2: I sent \$2.00, receiver returned \$0 (betrayal!)
- Round 3: Sender sent me \$0 (maybe retaliating? or cautious?)

The zero in Round 3 could be retaliation for my Round 2 behavior, or it might stem from communication distortion about what I actually returned in Round 1. Since each round likely involves different partners, the person who wit...

usage:

```
{
  "input_tokens": 937,
  "output_tokens": 851,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.36.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

content:

The sender sent \$0 this round, so I received \$0. With nothing received, I have nothing to return:

```
{'return': 0}
```

reasoning:

The sender sent \$0 this round, so I received \$0. I have nothing to return.

This is straightforward - I can only return 0 since I received nothing.

usage:

```
{
  "input_tokens": 1158,
  "output_tokens": 82,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.36.

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

content:

Looking at the pattern:

- Round 1: Good cooperation when I was receiver
- Round 2: I sent \$2.00, got little/nothing back
- Round 3: Received nothing as receiver

- Round 4: I sent \$1.50, got some return (earnings increased to \$11.36)
- Round 5: Received nothing as receiver again

Cooperation has been inconsistent, though the communication noise mentioned could be affecting perceptions on both sides. The receiver in Round 4 did return something when I sent , which is a positive signal.

I'll send a small amount to signal continued willingness to cooperate while being cautious :

```
{'send': 1.00}
```

#### reasoning:

Let me analyze the pattern of play:

Round 1: I was RECEIVER

- Sender sent \$2.05 (41%)
- I received \$6.15
- I returned \$2.50
- My earnings: \$3.65

Round 2: I was SENDER

- I sent \$2.00
- My total earnings after Round 2: \$6.65
- This means I gained \$3.00 in Round 2 ( $6.65 - 3.65 = 3.00$ )
- I started with \$5, sent \$2, left with \$3
- If I got back nothing, I'd have \$3 total, but I have \$6.65 total
- Actually wait, let me recalculate. After Round 1, I had \$3.65
- In Round 2 as sender, I get \$5 new endowment, send \$2, keep \$3
- So I should have  $3.65 + \$3 = \$6.65$  if I got nothing back
- My total is \$6.65, so I got \$0 back in Round 2

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- Total earnings stayed at \$6.65

Round 4: I was SENDER

- I sent \$1.50
- My total earnings after Round 4: \$6.65
- Wait, that doesn't make sense. Let me check again.
- Before Round 4, I had \$6.65
- In Round 4, I get new \$5 endowment, send \$1.50, keep \$3.50
- If I got nothing back, I'd have  $6.65 + \$3.50 = \$10.15$
- But my total after Round 4 should be reflected in what's stated before Round 5
- The prompt shows \$11.36 total before Round 5, so after Round 4 I had \$11.36
- That's a gain of \$4.71 from the \$6.65 I started with
- Since I kept \$3.50 from my endowment, the receiver must have returned \$1.21

Round 5: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My total earnings stayed at \$11.36

Now for Round 6: I'm SENDER again

- My total is still \$11.36

Looking at the pattern, cooperation has been sporadic. Round 1 showed genuine reciprocity when the sender trusted me with 41% and I returned generously. But Round 2 went nowhere- I sent 40% and got nothing back. Round 3 and 5 both had senders sending nothing. Round 4 was the only other time I sent (30%), and while I did get \$1.21 back, that's only 27% reciprocation on what they received- quite weak. The communication issues mentioned might be playing a role in this inconsistency...

#### usage:

```
{
  "input_tokens": 1237,
  "output_tokens": 1167,
  "reasoning_tokens": 0
}
```



**Message 8 role=user**

**content:**

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.75.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 9 role=assistant**

**content:**

The sender sent \$0 this round, so I received \$0. With nothing to return:

{'return': 0}

**reasoning:**

The sender sent \$0 this round, so I received \$0. I have nothing to return.

**usage:**

```
{
  "input_tokens": 1452,
  "output_tokens": 59,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$15.75.

How much do you send to the receiver this round? (0-5)

**Message 11 role=assistant**

**content:**

Looking at my earning progression:

- When I've been sender in recent rounds (4 & 6), I actually got good returns despite initial appearances
- Round 4: Sent \$1.50, earnings jumped significantly
- Round 6: Sent \$1.00, earnings increased again nicely
- When I'm receiver, senders send \$0

The pattern suggests that when I send, I'm getting cooperated with. I'll send a moderate amount to continue this potentially profitable pattern:

{'send': 2.00}

**reasoning:**

Let me review the pattern:

- Round 1: I was receiver, cooperated well
- Round 2: I was sender, sent \$2.00
- Round 3: I was receiver, got \$0
- Round 4: I was sender, sent \$1.50, got some return (earnings went from \$6.65 to \$11.36, so I got about \$4.71 back from that round)
- Round 5: I was receiver, got \$0
- Round 6: I was sender, sent \$1.00, got some return (earnings went from \$11.36 to \$15.75, so I got about \$4.39 back from that round)
- Round 7: I was receiver, got \$0

So when I'm the receiver, I'm getting \$0. But when I'm the sender, I'm actually getting returns. This is interesting.

Round 4: Started with \$6.65, sent \$1.50, so I should have  $\$6.65 - \$1.50 = \$5.15$ , plus whatever return I got. I ended at \$11.36, so return was about  $\$11.36 - \$5.15 = \$6.21$ . That's actually a very generous return!

Round 6: Started with \$11.36, sent \$1.00, so I should have  $\$11.36 - \$1.00 = \$10.36$ , plus whatever return I got. I ended at \$15.75, so return was about  $\$15.75 - \$10.36 = \$5.39$ . Also a generous return!

So when I send as sender, I'm getting very good returns. But when I'm receiver, the sender sends \$0.

Maybe the other player(s) are only cooperating when they're receivers, not when they're senders? Or maybe there's noise affecting what they see?

Given that I've been getting good returns when I send (despite what seemed like poor cooperation earlier), I should probably continue sending. ...

**usage:**

```
{
  "input_tokens": 1267,
  "output_tokens": 607,
  "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$20.37.

This round, The sender sent you \$0.21, so you received \$0.63. How much do you return to the sender? (0-0.63)

**Message 13 role=assistant**

**content:**

The sender sent me \$0.21, which tripled to \$0.63. This is the first time a sender has sent me something in several rounds!

I'll return a fair portion to encourage continued cooperation:

```
{'return': 0.30}
```

**reasoning:**

The sender sent me \$0.21, which was multiplied by 3, so I received \$0.63.

This is the first round where a sender has sent me something other than \$0! The amount is small (\$0.21), but it's a signal of cooperation.

They sent \$0.21 out of \$5, keeping \$4.79.

I received \$0.63.

I should return something to encourage cooperation. Since they sent a small amount, I should return a reasonable portion to signal I'm willing to cooperate.

If I return \$0.30, that's about half of what I received. That seems fair and reciprocal.

- Sender would end with:  $4.79 + 0.30 = 5.09$

- I would end with:  $0.63 - 0.30 = 0.33$

This is a positive signal that could build trust for future rounds.

Let me return \$0.30 to show I'm willing to reciprocate cooperation.

**usage:**

```
{
  "input_tokens": 1257,
  "output_tokens": 291,
  "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$20.7.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the pattern from when I'm sender:

- Round 4: Sent \$1.50, got a decent return
- Round 6: Sent \$1.00, got a return
- Round 8: Sent \$2.00, got a good return

As receiver in Round 9, I just returned \$0.30 out of \$0.63 to the sender - showing good faith cooperation.

The receivers have generally been returning something when I send, making it profitable. I'll send a bit more this round to signal stronger cooperation:

```
{'send': 2.50}
```

**Reasoning:**

Let me track my earnings:

- Started Round 4 at \$6.65
- Round 4: I sent \$1.50 as sender, ended at \$11.36 (gain of \$4.71)
- Round 5: I was receiver, got \$0, returned \$0, ended at \$11.36 (no change)
- Round 6: I sent \$1.00 as sender, ended at \$15.75 (gain of \$4.39)
- Round 7: I was receiver, got \$0, returned \$0, ended at \$15.75 (no change)
- Round 8: I sent \$2.00 as sender, ended at \$20.37 (gain of \$4.62)
- Round 9: I was receiver, got \$0.63, returned \$0.30, ended at \$20.7 (gain of \$0.33 from keeping \$0.33)

Pattern analysis:

- When I send as sender, I've been getting good returns! Each time I sent, my earnings increased substantially more than what I sent.
- When I'm receiver, senders usually send nothing (except round 9 where they sent \$0.21)
- In round 9, I received \$0.63 and returned \$0.30 (roughly half)

Looking at the numbers:

- Round 4: Sent \$1.50, gained \$4.71. This means I kept  $5 - 1.50 = 3.50$  + return. So return was  $4.71 - 3.50 = 1.21$
- Round 6: Sent \$1.00, gained \$4.39. This means I kept  $5 - 1.00 = 4.00$  + return. So return was  $4.39 - 4.00 = 0.39$
- Round 8: Sent \$2.00, gained \$4.62. This means I kept  $5 - 2.00 = 3.00$  + return. So return was  $4.62 - 3.00 = 1.62$

...

```
Usage: {
  "input_tokens": 1258,
  "output_tokens": 688,
  "reasoning_tokens": 0
}
```

#### GAME - Agent\_1 - Receiver - dyad\_1

Timestamp: 2026-08-12T09:54:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$6.66.

This round, The sender sent you \$0.98, so you received \$2.94. How much do you return to The sender? (0-2.94)

**Message 3 role=assistant**

**content:**

Looking at this situation:

The sender sent me \$0.98, which became \$2.94 after the 3x multiplier. This created value through cooperation.

To encourage continued cooperation and show trustworthiness, I should return a fair amount. Let me consider the gains:

- Total value created from cooperation: \$1.96 (\$2.94 - \$0.98)
- A fair return would split these gains reasonably

If I return \$1.50:

- Sender gains: \$5 - \$0.98 + \$1.50 = \$5.52 (net gain: \$0.52)
- I gain: \$2.94 - \$1.50 = \$1.44 (net gain: \$1.44)

This gives us both positive gains from cooperating, with me gaining slightly more since I received the multiplied amount.

```
'''json
{"return": 1.50}
'''
```

**reasoning:**

Let me analyze this situation:

The sender sent me \$0.98, which was multiplied by 3x to give me \$2.94.

I need to decide how much to return. Let me think about fairness and cooperation:

1. The sender gave up \$0.98 from their \$5 endowment
2. This created \$2.94 for me
3. The total value created is \$2.94 - \$0.98 = \$1.96 in additional value due to the multiplier

A fair split might consider:

- Equal split of gains: The sender sacrificed \$0.98, I got \$2.94. If I return nothing, sen

der has \$4.02 and I have \$2.94. If we split the \$1.96 gain equally, I should return enough so we each gain about \$0.98.

- To give the sender a total gain of ~\$0.98, they need:  $\$5 - \$0.98 + \text{return} = \$5.98$ , so return = \$0.98
- That seems too low. Let me think differently.

Alternative approach - equal final outcomes:

- Sender started with \$5, spent \$0.98, has \$4.02 + return
- I started with \$0, got \$2.94, keep \$2.94 - return
- For equal outcomes:  $\$4.02 + \text{return} = \$2.94 - \text{return}$
- $2 * \text{return} = \$2.94 - \$4.02 = -\$1.08$
- This gives negative, which doesn't make sense

Let me think about reciprocity:

- The sender trusted me with \$0.98
- This created value of \$1.96 (the difference between 3x multiplier and original)
- A reasonable return might be to give back enough that we both benefit from the cooperation
- Testing different return amounts: returning \$1.47 would give us nearly equal final gains (\$0.49 for them, \$1.47 for me), while \$1.50 provides them \$0.52 in gains. Even \$1.20 leaves them with only \$0.22 in gains while I keep \$1.74....

usage:

```
{
  "input_tokens": 938,
  "output_tokens": 1003,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$8.1.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at my experience so far:

From my earnings progression:

- Round 3: I sent \$2, but my earnings only increased by \$3 (the \$3 I kept), suggesting I got \$0 back
- Round 4: I received \$2.94 and returned \$1.50, showing goodwill

The receivers I've encountered have shown poor reciprocity (0% and 18% return rates, though communication noise may be a factor). However, these might be different people each round.

Given this pattern, I need to be more cautious while still leaving room for potential cooperation. The 3x multiplier means even modest returns can be profitable.

I'll send a smaller amount to limit risk while testing if this receiver will reciprocate:

```
'''json
{"send": 1}
'''
```

This sends \$1, creating \$3 for the receiver. If they return even \$1-1.50, we both benefit.

If they don't reciprocate, my loss is limited to \$1.

**reasoning:**

Let me track my earnings:

- Round 1: I sent \$3, which should have given me back something. My total after all rounds up to Round 2 was \$3.66. So I kept  $\$5 - \$3 = \$2$ , plus whatever was returned = \$3.66. So I received \$1.66 back from round 1.
- Round 2: I was receiver, sender sent \$0, so no change
- Round 3: I sent (need to calculate). Total after round 3 was still \$3.66? No wait, the prompt says "Your total earnings across all rounds are \$3.66" in Round 3. Let me recalculate.

Actually, let me track this more carefully:

- Round 1: I was sender, sent \$3. Total earnings after rounds 1-2 was \$3.66
- Round 2: I was receiver, got \$0. Total after rounds 1-2 was \$3.66

- Round 3: I was sender, sent \$2. Total after rounds 1-3 was \$6.66
- Round 4: I was receiver, got \$2.94, returned \$1.50. Total after rounds 1-4 was \$8.10

So in Round 3, I went from \$3.66 to \$6.66 = +\$3.00

I sent \$2, so I kept \$3 from my endowment

To gain \$3 total, I needed the return to be \$0

So the receiver in Round 3 returned \$0! That's disappointing.

In Round 4, the sender sent me \$0.98 (became \$2.94). I returned \$1.50. I gained \$2.94 - \$1.50 = \$1.44, which matches: \$6.66 + \$1.44 = \$8.10 ?

So my experience:

- Round 1: Sent \$3 (became \$9), got back \$1.66 (18% return)
- Round 3: Sent \$2 (became \$6), got back \$0 (0% return)
- Round 4: Received \$0.98 (became \$2.94), returned \$1.50

The reciprocity has been weak overall. Still, the 3x multiplier means even modest returns create value. I'm torn between sending nothing to protect myself and sending a small amount to test whether cooperation might work. If I send \$0, I get nothing back. A small send could still be worthwhile if receivers return even 20-30%...

usage:

```
{
  "input_tokens": 1195,
  "output_tokens": 992,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.91.

This round, The sender sent you \$0.36, so you received \$1.08. How much do you return to the sender? (0-1.08)

Message 7 role=assistant

content:

Looking at this situation:

The sender sent me \$0.36, which became \$1.08 after the 3x multiplier. This is a very small amount, suggesting they have low trust - likely from poor past experiences.

To demonstrate trustworthiness and potentially build cooperation:

Total value created from cooperation: \$1.08 - \$0.36 = \$0.72

If I return \$0.60:

- Sender gains: \$5 - \$0.36 + \$0.60 = \$5.24 (net +\$0.24)
- I gain: \$1.08 - \$0.60 = \$0.48 (net +\$0.48)

This shows strong reciprocity (~56% return rate) which could encourage more trust and larger exchanges in future rounds, benefiting both of us.

```
'''json
{"return": 0.60}
'''
```

reasoning:

Let me analyze this situation:

The sender sent me \$0.36, which became \$1.08 after the 3x multiplier.

This is a very small amount - they're being quite cautious, likely due to past experiences with low reciprocity.

Let me think about what's fair:

- Total value created: \$1.08 - \$0.36 = \$0.72
- A fair split of gains might be around half each

If I return \$0.54:

- Sender: \$5 - \$0.36 + \$0.54 = \$5.18 (gain of \$0.18)
- Me: \$1.08 - \$0.54 = \$0.54 (gain of \$0.54)

This gives us both positive gains, with me gaining more since I received the multiplied amount.

Alternatively, if I return \$0.70:

- Sender:  $\$5 - \$0.36 + \$0.70 = \$5.34$  (gain of \$0.34)
- Me:  $\$1.08 - \$0.70 = \$0.38$  (gain of \$0.38)

This is closer to an equal split of the total value created.

Given that:

1. The sender sent a very small amount (showing low trust)
2. I want to demonstrate trustworthiness to encourage future cooperation
3. Even a modest return shows good faith
4. The amounts are small, so being generous could pay off

I'll return about 50-65% of what I received, which shows strong reciprocity and might encourage the sender to trust more in future rounds.

Let me return \$0.60:

...

**usage:**

```
{
  "input_tokens": 1485,
  "output_tokens": 669,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.39.

How much do you send to the receiver this round? (0-5)

**Message 9 role=assistant**

**content:**

Looking at my track record:

As SENDER:

- Round 1: Sent \$3, got back \$1.66 (low reciprocity)
- Round 3: Sent \$2, got back \$0 (no reciprocity)
- Round 5: Sent \$1, got back \$0.81 (best reciprocity rate ~27%)

As RECEIVER, I've been returning 50-56%, demonstrating trustworthiness.

Interestingly, my smallest send (Round 5) got the best return rate. This could be due to:

- Communication noise making small amounts appear larger
- Building reputation from my fair behavior as receiver
- Different receivers with varying trustworthiness

I'll send a moderate amount to test if cooperation is improving:

```
'''json
{"send": 2}
'''
```

This sends \$2 (becomes \$6). If reciprocity continues improving, we both benefit. If not, my loss is limited.

**reasoning:**

Let me track my earnings:

- Round 1 (SENDER): I sent \$3, ended with \$3.66. So I kept \$2 and got back \$1.66
- Round 2 (RECEIVER): Sender sent \$0, I got \$0. No change to earnings.
- Round 3 (SENDER): Started at \$3.66. I sent \$2. Ended at \$6.66, so gained \$3. This means I kept \$3 and got back \$0.
- Round 4 (RECEIVER): Started at \$6.66. Received \$2.94, returned \$1.50. Gained \$1.44 (\$2.94-\$1.50). Ended at \$8.1.
- Round 5 (SENDER): Started at \$8.1. I sent \$1. Ended at \$12.91, so gained \$4.81. I kept \$4, so I got back \$0.81.
- Round 6 (RECEIVER): Started at \$12.91. Received \$1.08, returned \$0.60. Gained \$0.48 (\$1.08-\$0.60). Ended at \$13.39.
- Round 7 (SENDER): Starting at \$13.39

Looking at patterns when I was SENDER:

- Round 1: Sent \$3, got back \$1.66 (18.4% of the \$9 received)
- Round 3: Sent \$2, got back \$0 (0% of the \$6 received)
- Round 5: Sent \$1, got back \$0.81 (27% of the \$3 received)

Interesting - Round 5 showed better reciprocity (27% return) when I sent less. And I've been showing good reciprocity when I'm a receiver (50% in Round 4, 56% in Round 6).

The pattern suggests:

1. Different receivers each round

...

**usage:**

```
{
  "input_tokens": 1735,
  "output_tokens": 797,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$17.240000000000002.

This round, The sender sent you \$0.99, so you received \$2.9699999999999998. How much do you return to The sender? (0-2.9699999999999998)

**Message 11 role=assistant**

**content:**

Looking at this situation:

The sender sent me \$0.99, which became approximately \$2.97 after the 3x multiplier.

I've been maintaining a consistent pattern of reciprocity as a receiver (50-56% return rates in previous rounds), which helps build trust and encourages future cooperation.

For this round:

- Value created from cooperation:  $\$2.97 - \$0.99 = \$1.98$

If I return \$1.50:

- Sender gains:  $\$5 - \$0.99 + \$1.50 = \$5.51$  (net +\$0.51)
- I gain:  $\$2.97 - \$1.50 = \$1.47$  (net +\$1.47)

This maintains my fair reciprocity pattern (~50% return rate) and ensures we both benefit from cooperation.

```
'''json
{"return": 1.50}
'''
```

**reasoning:**

The sender sent me \$0.99, which became approximately \$2.97 after the 3x multiplier.

Looking at what I've been doing as a receiver:

- Round 2: Received \$0, returned \$0 (nothing to return)
- Round 4: Received \$2.94, returned \$1.50 (~51% return rate)
- Round 6: Received \$1.08, returned \$0.60 (~56% return rate)

I've been consistently fair as a receiver, returning about 50-56% of what I receive. This helps build trust and encourages cooperation.

For this round:

- Sender sent: \$0.99
- I received: ~\$2.97
- Value created:  $\$2.97 - \$0.99 = \$1.98$

A fair return would be around 50-55% of what I received, which would be \$1.48-\$1.63.

If I return \$1.50:

- Sender gets:  $\$5 - \$0.99 + \$1.50 = \$5.51$  (net gain: \$0.51)
- I get:  $\$2.97 - \$1.50 = \$1.47$  (net gain: \$1.47)

This maintains my pattern of reciprocity and shares the gains fairly, with me getting slight



htly more since I received the multiplied amount.

**usage:**

```
{
  "input_tokens": 1859,
  "output_tokens": 535,
  "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$18.71.

How much do you send to the receiver this round? (0-5)

**Message 13 role=assistant**

**content:**

Looking at my track record:

As SENDER:

- Round 1: Sent \$3 ? got \$1.66 back (18% return)
- Round 3: Sent \$2 ? got \$0 back (0% return)
- Round 5: Sent \$1 ? got \$0.81 back (27% return - best!)
- Round 7: Sent \$2 ? got \$0.85 back (14% return)

Pattern: Despite my fair behavior as a receiver (50-56% returns), receivers have consistently shown poor reciprocity. Interestingly, my smallest send (\$1) yielded the best return rate.

Given this pattern and considering communication noise may be affecting perceptions, I'll send a smaller amount:

```
'''json
{"send": 1}
'''
```

This limits my downside risk to \$1 while still creating \$3 for the receiver. If they reciprocate even modestly, we both gain. Based on past rounds, smaller sends seem to perform better.

**reasoning:**

Let me track my earnings:

- Started Round 3 with \$3.66
- Round 3 (SENDER): Sent \$2, kept \$3, got back ~\$0 ? total \$6.66
- Round 4 (RECEIVER): Got \$2.94, returned \$1.50, kept \$1.44 ? total \$8.10
- Round 5 (SENDER): Sent \$1, kept \$4, got back \$0.81 ? total \$12.91
- Round 6 (RECEIVER): Got \$1.08, returned \$0.60, kept \$0.48 ? total \$13.39
- Round 7 (SENDER): Sent \$2, kept \$3, got back ???
  - Started with \$13.39, now at \$17.24, so gained \$3.85
  - That means I got back \$0.85 (kept \$3 + returned \$0.85 = \$3.85)
- Round 8 (RECEIVER): Got \$2.97, returned \$1.50, kept \$1.47 ? total \$18.71

So in Round 7, I sent \$2 and got back only \$0.85 (return rate ~14%).

My experience as SENDER:

- Round 1: Sent \$3 ? got \$1.66 back (return rate 18%)
- Round 3: Sent \$2 ? got \$0 back (return rate 0%)
- Round 5: Sent \$1 ? got \$0.81 back (return rate 27%)
- Round 7: Sent \$2 ? got \$0.85 back (return rate 14%)

The pattern shows:

- Receivers are generally not reciprocating well
- My smaller sends (\$1) actually got better percentage returns
- Even though I've been a fair receiver (returning 50-56%), this hasn't improved sender experiences

Given this consistent poor reciprocity from receivers, and considering communication noise may be affecting things, I should:

...

**usage:**

```
{
  "input_tokens": 1997,
  "output_tokens": 746,
}
```

```

    "reasoning_tokens": 0
  }
Message 14 role=user
  content:
    Round 10

    You are the RECEIVER this round. Your total earnings across all rounds are $22.71.
    This round, The sender sent you $0, so you received $0. How much do you return to The sender? (0-0)
Assistant output:
  Looking at this situation:

  The sender sent me $0, so I received $0 after the multiplier.

  Since I received nothing, there's nothing to return.

  '''json
  {"return": 0}
  '''
Reasoning:
  The sender sent me $0, so I received $0 after the 3x multiplier (which is still $0).

  Since I received nothing, I have nothing to return. The only valid response is to return $0.

  This is straightforward - there's no trust game happening this round since the sender chose not to send anything.
Usage: {
  "input_tokens": 2038,
  "output_tokens": 131,
  "reasoning_tokens": 0
}

```